



Forecasting to support EMS tactical planning: what is important and what is not

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Abstract

Forecasting emergency medical service (EMS) call volumes is critical for resource allocation and planning. The development of many commercial and free software packages has made a variety of forecasting methods accessible. Practitioners, however, are left with little guidance on selecting the most appropriate method for their needs. Using 5 years of data from 3 cities in Alberta, we compute exponential smoothing and benchmark forecasts for 8-hour periods for each ambulance station catchment area and with a forecast horizon of two weeks—a spatio-temporal resolution appropriate for tactical planning. The methods that we consider differ on three spectra: the number and type of time-series components, whether forecasts are computed individually or jointly, and the way in which forecasts at a specific resolution are converted to forecasts at the resolution of interest. We find that it is important to include a weekly seasonal component when forecasting EMS demand. Multiplicative seasonality, however, shows no benefit over additive seasonality. Adding other time-series components (e.g., trend, ARMA errors, Box-Cox transformation) does not improve performance. Spatial resolutions of station catchment area and lower, and temporal resolution of 4–24 hours perform similarly. We adapt an existing hierarchical forecasting framework to a two-dimensional spatio-temporal hierarchy, but find that hierarchical reconciliation of forecasts does not improve performance at the forecast resolution of interest for tactical planning. Neither does jointly forecasting time series. We show that added complexity does not materially improve forecasting performance. The simple methods that we find perform well are easy to implement and interpret, making implementation in practice more likely. In a simulation study we alter the empirical weekly patterns and demonstrate how extreme differences between the weekly seasonality patterns of different regions cause hierarchically-reconciled bottom-up approaches to outperform top-down approaches.

Keywords Empirical research · Health care management · Service operations · Emergency medical services · Hierarchical forecasting · Operations research · Operations management

Highlights

- We use exponential smoothing methods to forecast EMS call volumes for 8-hour periods and station catchment areas; a spatio-temporal resolution appropriate for tactical planning
- We find that it is important to include a weekly seasonal component, but adding other time-series components (e.g., trend, ARMA errors, Box-Cox transformation) does not improve performance. Neither does jointly forecasting time series.
- Simple top-down approaches that use constant proportions to divide a daily forecast for an entire city into 8-hour periods and station catchments areas perform very well.

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1 Introduction

An effective healthcare system is crucial for ensuring the wellbeing of citizens. Emergency medical services (EMS) play a critical role as the first point of entry into the system

for many patients. Careful planning is necessary to ensure that resources are available when and where they are most needed. Reliable forecasts are a necessary starting point for all planning. We focus on forecasts to support tactical planning activities, such as assignment of vehicles and paramedic crews to stations and scheduling of paramedic shifts.

Urban EMS systems with many stations need to compute many forecasts to support tactical planning and few of them employ forecasting experts, which makes simple methods preferred. Modern exponential smoothing (ES) methods are relatively simple, yet they regularly outperform more complicated methods in forecasting competitions [1] and therefore we focus on ES methods. Consistent with our focus on tactical planning, we provide forecasts for ambulance station catchment areas (to support assignment of vehicles and crews to stations) and 8-hour periods (a typical shift length), for forecast horizons of up to 2 weeks, a typical leadtime for changing shift schedules [2, 3]. The methods differ on two spectra: the number and type of time-series components (for example, additive/multiplicative trend or seasonality) and the way in which they forecast for the desired resolution using forecasts at higher or lower resolutions. We use *top-down* approaches to divide forecasts to achieve higher resolution, and *bottom-up* approaches to combine forecasts to achieve lower resolution, as needed. We also consider *hierarchical forecasts* that use forecasts at many different spatio-temporal resolutions for coming up with *reconciled* forecasts at the resolution of interest.

We pay special attention to three challenges. First, for greater generalizability, we test performance on data from multiple cities. This is important because spatio-temporal patterns of EMS calls vary with city characteristics such as its geography, its infrastructural layout, and the size and distribution of the population [4]. To address this challenge, we use five years of data for three cities in Alberta, Canada: Calgary, Edmonton, and Grande Prairie—a more diverse data set than has been used in past research. Table 1 compares the three cities. Calgary and Edmonton are the two largest cities in Alberta, each with a population exceeding 900 thousand. Grande Prairie is smaller, with a population of 63 thousand. Calgary and Edmonton are geographically similar in that both have a river that runs through the city center (Fig. 1). This diverse data set allows us to investigate whether a method that performs well in one city also performs well in other cities with similar or different characteristics.

Second, a critical decision when designing an EMS forecasting system is choosing the appropriate spatio-temporal resolution. From a technical point of view, increasing resolution, by predicting for shorter time intervals and smaller regions, results in fewer calls per interval/region which can lead to noisier forecasts. Decreasing resolution, on the other hand, could cause analysts to miss spatio-temporal patterns that are evident at higher resolutions and could also result in

aggregation error [5, 6]. In order to investigate this trade-off, we use three levels of temporal granularity: 4-hour, 8-hour, and 24-hour periods, and three levels of spatial granularity: station catchment areas, intermediate regions, and the whole city. This results in nine combinations, which we will use to train different methods. These values were selected after interviewing EMS system managers and discussing the ways in which forecasts might be used for tactical planning. For example, in order to optimize the fleet size of an ambulance station, we need to know the demand for the station's catchment area. Having forecasts for smaller regions than catchment areas does not provide any further advantage, as long as all ambulances are deployed from stations. As another example, if permissible paramedic shift start times are midnight, 8 am, and 4 pm, then demand forecasts for 8-hour periods would be sufficient for staff planning. Some past research has used higher spatial resolution [7]. This could be useful for selecting station locations (a strategic planning activity) but it is not clear how this finer granularity is useful for tactical planning of deployment of ambulances and crews to existing stations. Since the focus of this paper is on tactical as opposed to strategic planning we do not consider higher spatial resolution than station catchment areas.

Third, we compare methods to compute forecasts at a desired resolution, using forecasts at other resolutions, including top-down, bottom-up, and hierarchical approaches to forecasting. A top-down approach divides a low spatial resolution forecast to obtain forecasts for smaller regions. A bottom-up approach aggregates high spatial resolution forecasts to obtain forecasts for larger regions. Hierarchical forecasting involves reconciling forecasts from many related time series that can be organized within a hierarchical structure. In addition to comparing the performance of these approaches on our empirical data, we perform additional comparisons using simulated data, to better understand the drivers of performance differences.

We focus on ES methods because they are relatively simple to use and have been shown to perform well in practice in a variety of settings [1]. Furthermore, the innovations state-space model framework that we use encompasses many other methods (for example, all ARIMA models) as special cases. We include three benchmark methods for comparison: MEDIC (see [7]), TBATS (see [8]) and a multivariate method by [9], which we refer to as MV. We do not expect to find a single method that performs best for all regions and all time intervals. Rather, we hope to identify one or a few methods that perform well (say, within 5% of the best method) for most regions and most time intervals.

The rest of the paper is organized as follows. Section 2 reviews related literature. Section 3 describes our data and preprocessing. Section 4 defines notation and provides descriptive statistics. Section 5 discusses the methods that we compare. Section 6 shows how we evaluate performance.

Table 1 Population, area, and call volume for the cities in our sample (2016 values)

City	Population	Area (km ²)	Pop. dens. (per km ²)	Total calls	Calls per capita	Number of stations
Calgary	1,239,220	825.56	1,501	99,104	1/12.5	27
Edmonton	932,550	685.25	1,361	89,286	1/10.4	16
Grande Prairie	63,166	132.72	476	4,573	1/13.8	1

Section 7 reports on a simulation study to further investigate top-down and bottom-up approaches. Section 8 reports results, based on out-of-sample performance. Section 9 compares how the top performing methods compute their forecasts. Section 10 concludes.

2 Literature review

We survey four streams of related literature: temporal time-series forecasting of EMS call volume, spatio-temporal time-series forecasting of EMS call volume, forecasting methods that use additional predictors, and interpretable machine learning.

2.1 Temporal time-series forecasting of EMS call volume

This stream focuses on computing a single set of forecasts for a geographic region (such as a city), using only time-series data. The MEDIC moving-average method is commonly used in the EMS industry [7]. The temporal resolution has increased over time, from annual [10], to daily [11–13], to hourly [12, 14–16]. A wide variety of methods have been

employed, including linear regression [10, 12], ARIMA [12–14], ES [11], and matrix decomposition [13, 16]. None of these studies provide a thorough comparison of possible forecasting methods. We compare a wide spectrum of relatively simple ES methods to two complex benchmark methods and find that the simple methods perform as well as the complex methods.

Call-center demand forecasting has been studied extensively; see [17] for a thorough overview. EMS demand forecasting is a type of call-center demand forecasting, with EMS calls typically received at a “public safety answering point” [18], such as a 911 call center. A distinguishing feature of EMS operations is that forecasting the spatial distribution of call locations is crucial, in order to support EMS planning. One way to forecast the spatial distribution is to separate the incoming calls into multiple streams, corresponding to different geographical regions. Non-EMS call centers also separate incoming calls into multiple streams that correspond to such features as the language that the caller speaks or the type of product for which the caller requires assistance. If these call streams depend on each other in some fashion, then joint multivariate forecasting of the corresponding time series, as in the MV method [9], could be beneficial. We find that the MV method performs no better than much simpler ES methods.

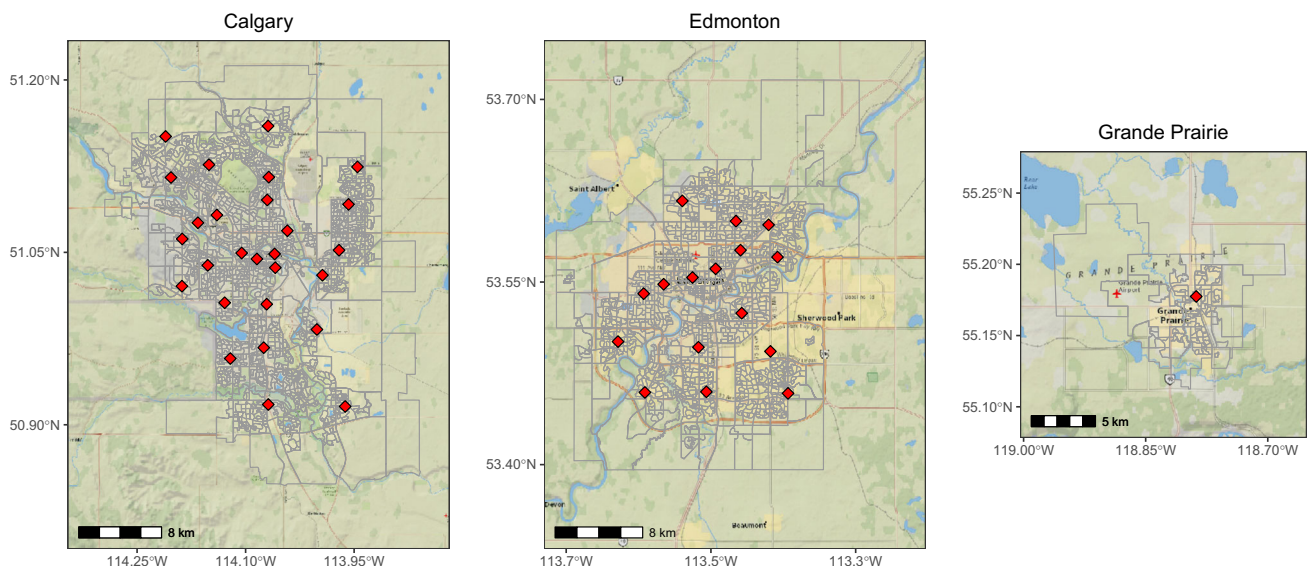


Fig. 1 Maps of Calgary, Edmonton, and Grande Prairie. Ambulance stations are shown as red diamonds. The polygons are dissemination areas, each having a population of 400–700

2.2 Spatio-temporal time-series forecasting of EMS call volume

This stream focuses on computing separate forecasts for subregions (such as neighborhoods in a city) using only time-series data. The spatial aspect has been studied less than the temporal. Existing studies of the spatial dimension have used a variety of methods, including mixtures of Gaussians and kernel wrapping for estimating spatial density functions [4, 19], neural networks, support vector regression, decision trees, and linear regression [7, 20]. The MV method, which we use as a benchmark, has not been previously used for spatio-temporal forecasting of EMS demand. Similar to the MV method, implementation and routine use is challenging for many of the spatio-temporal methods, requiring significant effort and technical expertise. It is therefore of interest to investigate whether these complex methods result in sufficient improvement in accuracy to justify the cost of using them and that is our focus.

The sparsity of EMS call volume makes high-resolution spatial forecasting challenging because many areas will have no calls for long time periods [7, 19]. Top-down approaches, which first forecast for a larger region and then divide the forecast into smaller regions, have been proposed as a way of dealing with sparse data [21]. Bottom-up approaches do the opposite: they aggregate forecasts for small regions to obtain forecasts for larger regions. Hierarchical forecasting [22–24] frameworks provide methods to obtain reconciled bottom-up forecasts for various types of hierarchies. We adapt such frameworks to a spatio-temporal hierarchy that is appropriate for EMS demand.

2.3 Forecasting methods that use additional predictors

This stream includes studies that combine other sources of information with historical call volumes for forecasting EMS demand. The effect of demographic factors has been studied since the early 1980s [25–29]. Typically, such studies focus on long-term forecasting, which can support strategic decisions, such as the selection of locations for new stations. We do not use demographic data because our focus is on short-term forecasting over time horizons that are too short for measurable changes to a region's demographics.

The impact of weather on EMS demand has been studied, and call volumes have been found to be correlated with current weather conditions [30, 31]. In order for weather information to be used for the routine calculation of forecasts to support tactical planning, weather forecasts or past weather information needs to be available at the time when the forecasts are calculated. Using accurate weather forecasts has been shown to improve accuracy [32], but using noisy

weather forecasts or lagged actual weather conditions has not been shown to improve accuracy [32, 33].

2.4 Interpretable machine learning

We conclude this section by pointing out that our work is in the spirit of interpretable machine learning [34, 35]. The forecasting equations for the best-performing ES methods are much easier to understand than for the benchmark methods, as we illustrate in Section 9. Interpretable methods are easier to troubleshoot, easier to use, and more trusted [35]. The creation of certain interpretable models requires the solution of computationally difficult optimization problems [35]. In contrast, the training of ES methods is straightforward.

3 Data description and preprocessing

We have data on the time and perturbed location (see Appendix A for details) of every EMS call received over 5 years, for Calgary and Edmonton, and over 4 years and 2 months, for Grande Prairie. Table 2 lists the number of calls received each year in each city.

We train methods with three levels of temporal resolution: 4, 8, and 24-hour intervals and three levels of spatial resolution: station catchment area, intermediate regions, and city-wide. Grande Prairie has only one station, and therefore we use only one spatial resolution for that city.

For a temporal resolution δ , starting at midnight, we divide the day into equally spaced intervals of length δ and count the number of calls in each interval. For example, for an 8-hour temporal resolution, we use midnight to 8 am, 8 am to 4 pm, and 4 pm to midnight.

To spatially aggregate calls, we begin by mapping each call to its dissemination area (DA; the smallest geographical areas in Canada for which full census data is available, [36]) and we use DAs as the building blocks for further aggregation.

We construct station catchment areas (Fig. 2,(a) and (c)) by assigning each DA to a station. We calculate the center of a DA as the population-weighted average of the dissemination blocks within that DA (A dissemination block is the smallest geographic area in Canada for which population figures are disseminated, [36].) We use the Google Maps API to calculate the travel time from all stations to the DA's center and assign the DA to the closest station.

The intermediate regions (Fig. 2,(b) and (d)) were constructed by combining station catchment areas, using geographical barriers (such as rivers), main roads and highways as boundaries. The intermediate regions have no official meaning—we constructed them to provide an additional level of comparison in this research.

Table 2 EMS calls by city and year

	2012 Apr-Dec	2013 Jan-Dec	2014 Jan-Dec	2015 Jan-Dec	2016 Jan-Dec	2017 Jan-Mar	Total
Calgary	64,744	89,551	93,994	95,843	99,104	25,199	468,435
Edmonton	58,283	80,359	84,263	87,754	89,286	22,613	422,558
Grande Prairie		Feb-Dec 4,005	4,629	4,777	4,573	1,253	19,237

We do not expect the perturbation of call locations to affect our findings, because the average perturbation area is roughly 400 m^2 , which is much smaller than the station catchment areas that we use for the highest spatial resolution. (The average station catchment areas are 31.4 , 43.8 , and 133 km^2 for Calgary, Edmonton, and Grande Prairie, respectively.)

4 Notation and preliminary data analysis

Let $y_{t,r}^{\delta,\alpha}$ be the call volume in Period t and Region r , using temporal resolution δ and spatial resolution α , where $t \in \{1, 2, \dots\}$ and $r \in R(\alpha)$. We focus on temporal resolutions of $\delta = 4, 8$, and 24 hours, but we also use $\delta = 1$ and 168 hours in this section, to describe weekly patterns. The spatial resolution can be $\alpha = H, M$, or L , short for *High* (station catchment areas), *Medium* (intermediate regions), or *Low* (whole city). The set $R(M)$, as an example, for Edmonton, is {Region 1, ..., Region 4}. We use $m_\alpha \equiv |R(\alpha)|$ to denote the number of regions at the α spatial resolution level. We omit the superscripts δ and α when they are not needed, so that the notation simplifies to $y_{t,r}$. We denote the call volume forecast for Period $t + h$ made in Period t as $\hat{y}_{t+h,r|t}$.

We define three functions:

- $f(t, \delta)$ gives the time interval of t , for example, $f(1, 8) = [0, 8)$ of Week 1.
- $g(t)$ gives a number from 1 to 21, which specifies the 8-hour interval of the week that Period t belongs to, with $1 = [0, 8)$, $2 = [8, 16)$, ..., $21 = [160, 168)$.
- $\omega(t)$ specifies the day of the week that Period t belongs to.

We use bold lower case for vectors and bold upper case for matrices, for example, $\mathbf{y}_t = (y_{t,r}, r \in R)$.

4.1 Temporal patterns

Figure 3 shows the percentage of calls for the 168 hours of the week, starting at 0 am on Monday, for the three cities. The black curve represents the average among all weeks in the data and the grey shades represent 95% confidence intervals. The procedure for creating these plots is:

- Calculate the proportion of calls in Hour h of Week j , of the n weeks of data (this detrends the data):

$$p_{j,h} = \frac{y_{168(j-1)+h}^{1,L}}{y_j^{168,L}}, \quad h = 1, \dots, 168, \quad j = 1, \dots, n$$



Fig. 2 (a) and (c): Ambulance station catchment areas (numbers indicate station locations); (b) and (d): Intermediate regions

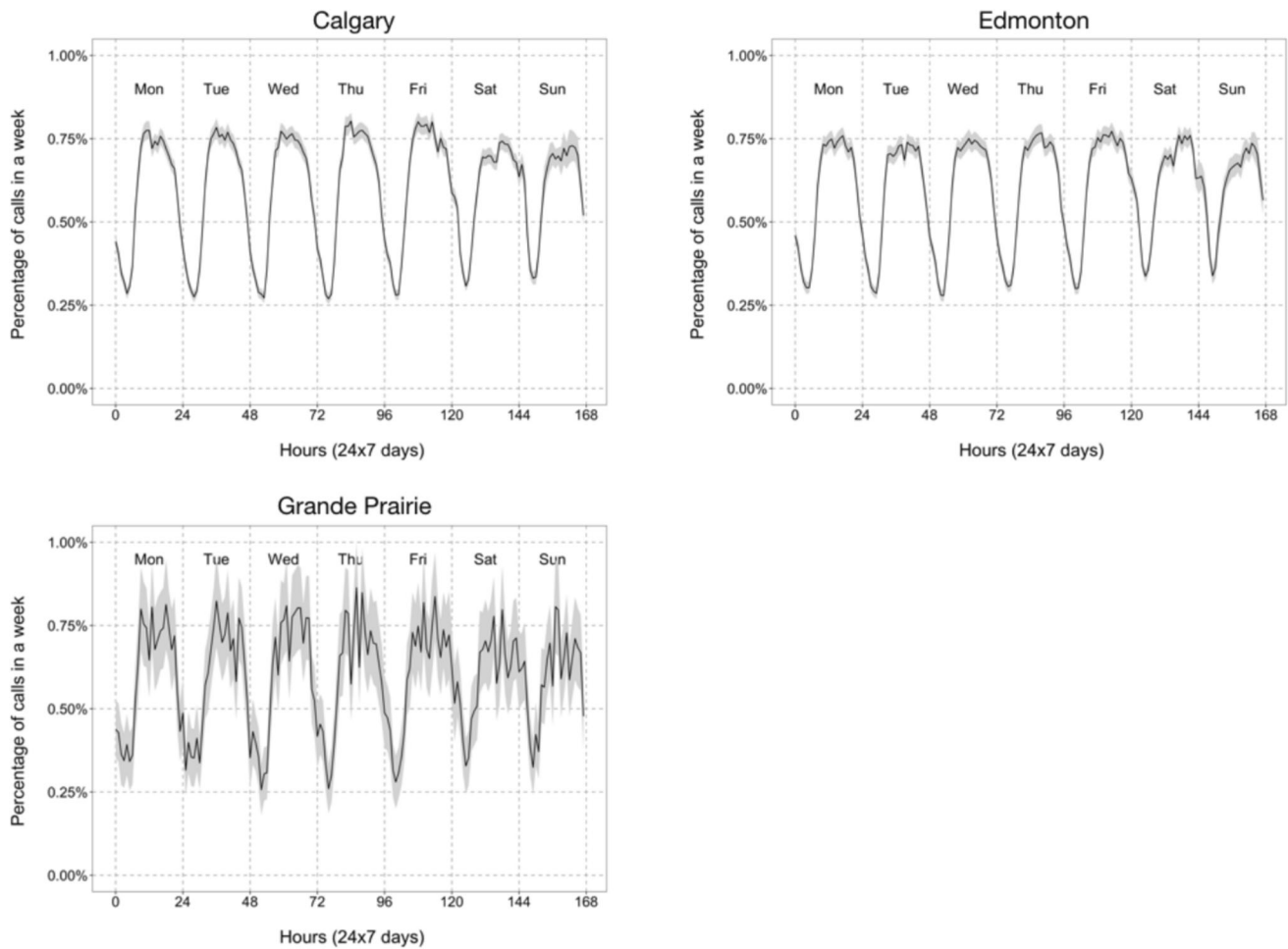


Fig. 3 Weekly temporal patterns of the detrended time series of call volumes

- Calculate the sample mean and sample variance of the $p_{j,h}$ s for each hour:

$$\mu_h = \frac{1}{n} \sum_{j=1}^n p_{j,h}; \quad \sigma_h^2 = \frac{1}{n-1} \sum_{j=1}^n (p_{j,h} - \mu_h)^2 \quad h = 1, \dots, 168$$

- Calculate a confidence interval half-width for each hour of the week: $t_{n-1}^{0.95} \sqrt{\frac{\sigma_h^2}{n}}$, where $t_{n-1}^{0.95}$ is the t-score, with $n - 1$ degrees of freedom, at 95% confidence level.

Each day, the call volume rises sharply from 5 am (its minimum) to 10 am and remains high until 8 pm. It then begins to fall and reaches a minimum again at 5 am the next day. The temporal patterns for the first four days of the week (Monday to Thursday) are quite similar and different from the weekends. In particular, call volumes on Fridays and Saturdays remain high for a longer period of time. This pattern is typical for EMS call volumes [37].

Grande Prairie, being a smaller city, has a noisier curve with wider confidence intervals than the other two cities. Calgary and Edmonton are similar, with the former having slightly higher peaks and deeper valleys.

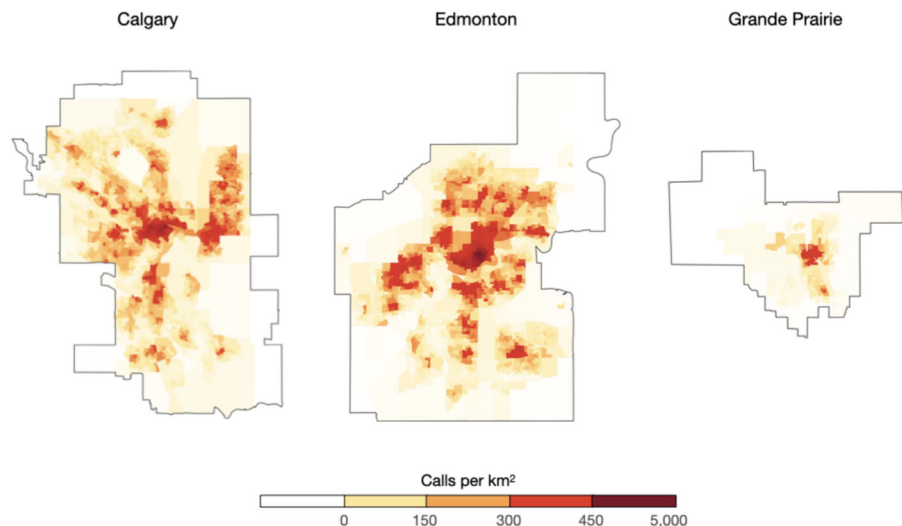
4.2 Spatial patterns

Figure 4 shows the spatial distribution of calls, with each DA coloured based on the number of calls per km² in 2016. The central business district has, by far, the highest concentration of calls in all three cities.

4.3 Spatio-temporal patterns

The spatial patterns and the temporal patterns interact, as we illustrate for Calgary in Fig. 5. Although the intermediate regions have similar weekly temporal patterns, we observe small but systematic differences, the most pronounced being the elevated call volumes after midnight on Friday and Sat-

Fig. 4 Spatial pattern of call volumes for 2016



urday in Region 1 (the central business district). We also see an earlier and higher peak on weekdays in Region 5 than for the city as a whole.

4.4 Preliminary estimates for trend and seasonality

The annual call volume increases from year to year, for all three cities, for all complete years, with one exception (Table 2). This year-to-year increase is especially noticeable for Calgary and Edmonton, possibly due to rapid population increase. Such patterns are often accounted for by including a trend component in the time series model.

We use measures from [38] to obtain preliminary assessments of the strength of the seasonal (S_t) and trend (T_t) components of the time series. These measures are obtained from a decomposition of the time series, such as the TSA method [39], implemented in the stats R package:

$$y_t = S_t + T_t + R_t,$$

where R_t is remainder component. The measures of the strength of trend and strength of seasonality are:

$$F_T = \max \left(0, 1 - \frac{\text{Var}(R_t)}{\text{Var}(T_t + R_t)} \right),$$

$$F_S = \max \left(0, 1 - \frac{\text{Var}(R_t)}{\text{Var}(S_t + R_t)} \right).$$

Both measures range from 0 to 1. A strongly trended time series should have much greater variation in the seasonally adjusted series, $T_t + R_t$, than in the remainder component, R_t , resulting in F_T close to 1; conversely, a weak trend component should result in F_T close to 0.

Table 3 shows F_T and F_S for Calgary, Edmonton, and Grande Prairie, using the 8-hr, city-wide time series, $\{y_t^{8,L}\}$. Calgary and Edmonton have $F_T > 0.5$ and $F_S > 0.85$, suggesting a relatively strong trend and a very strong seasonal pattern. Therefore, we include methods with trend and seasonal components in our comparison of forecasting methods.

Fig. 5 Interaction between spatial and temporal patterns, for Calgary. The graphs on the right show the weekly temporal pattern for each intermediate region (bold lines), superimposed on the weekly temporal pattern for the city as a whole (gray lines)

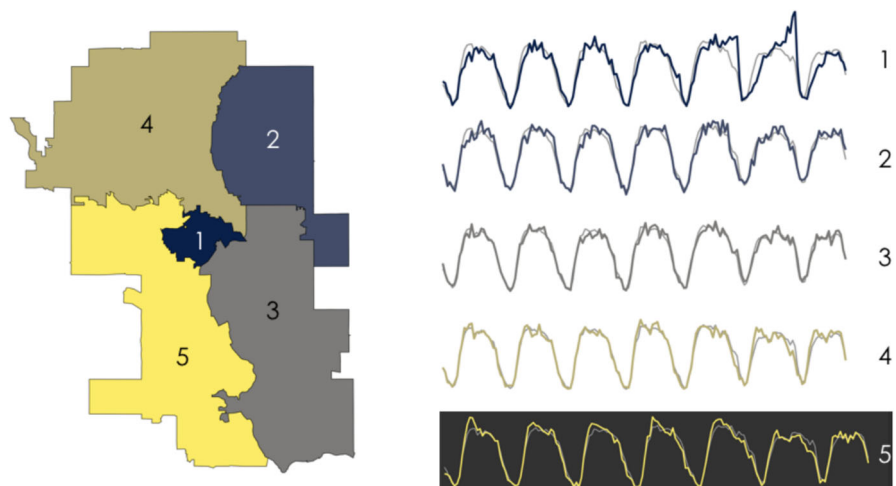


Table 3 Preliminary measures of trend and seasonality for the 8-hr, city-wide time series

	Calgary	Edmonton	Grande Prairie
F_T = Strength of trend	0.54	0.53	0.36
F_S = Strength of seasonality	0.88	0.86	0.32

4.5 Frequency of zero counts

The frequency of zero counts for the highest spatio-temporal resolution (4-hour periods, station catchment areas) is 28.8% for Calgary, 15.7% for Edmonton, and 14.8% for Grande Prairie. Frequent zeros can be problematic for estimation, assessment of fit, and calculation of scaled error measures, such as mean absolute percent error (MAPE). These difficulties provide a further reason for not using high spatio-temporal resolutions.

5 Models, methods, and estimation

Following [40] we distinguish between models and methods, using *model* for equations that provide an entire distribution of forecasts and *method* for equations or algorithms that only give point forecasts.

5.1 Univariate and multivariate models

We compare multiple variants of univariate ES methods, as well as the TBATS generalization of ES [8], which we include as a univariate benchmark. Innovations state-space models provide a unified way to formulate models corresponding to all of these methods. In addition, we describe the MEDIC industry-standard benchmark method and the MV multivariate benchmark method [9], and we use a hierarchical approach to reconcile univariate forecasts, as discussed in Section 5.2.

Innovations state-space models and ES:

The governing equations of a linear innovations state-space model are:

$$y_{t,r} = \mathbf{w}_r^T \mathbf{x}_{t-1,r} + \epsilon_{t,r}, \quad (1)$$

$$\mathbf{x}_{t,r} = \mathbf{F}_r \mathbf{x}_{t-1,r} + \mathbf{g}_r \epsilon_{t,r}, \quad (2)$$

where $\epsilon_{t,r}$ is white noise; $\mathbf{x}_{t,r}$ is the $k \times 1$ state vector; and the $k \times k$ matrix $\mathbf{F}_r$ and the $k \times 1$ vectors $\mathbf{g}_r$ and $\mathbf{w}_r$ contain coefficients. The *measurement equation* (1) specifies how the state vector translates into a call volume. The *transition equation* (2) describes how the state vector evolves. Most ES methods can be derived from one or more (linear or nonlinear)

innovations state space models. For example, consider the additive-seasonality variant of the Holt-Winters method [41, 42], whose point forecasts are generated as follows:

$$l_{t,r} = \alpha_r (y_{t,r} - s_{t-m,r}) + (1 - \alpha_r)(l_{t-1,r} + b_{t-1,r}), \quad (3)$$

$$b_{t,r} = \beta_r (l_{t,r} - l_{t-1,r}) + (1 - \beta_r)b_{t-1,r}, \quad (4)$$

$$s_{t,r} = \gamma_r (y_{t,r} - l_{t-1,r} - b_{t-1,r}) + (1 - \gamma_r)s_{t-m,r}, \quad (5)$$

$$\hat{y}_{t+h,r|t} = l_{t,r} + b_{t,r}h + s_{t-m+h_m^+,r}, \quad (6)$$

where $l_{t,r}$, $b_{t,r}$, and $s_{t,r}$ are the level, trend, and seasonality in Period t for Region r , m is the length of the seasonal cycle; $h_m^+ = [(h - 1) \bmod m] + 1$; and α_r , β_r , and γ_r are smoothing parameters. Equations 3–6 can be derived from a linear innovations state space model of the form shown in Eqs. 1–2 and also from a nonlinear innovations state space model in which the error term ϵ_t is multiplied with the state vector $\mathbf{x}_{t-1}$. In other words, a forecasting method can correspond to more than one model. The class of innovations state-space models includes models for all ES methods and models equivalent to all ARIMA models [40].

Hyndman et al [40] introduced a unified notation for innovations state-space models associated with ES methods, where the triple (E, T, S) identifies the type of error, trend, and seasonality. Table 4 uses ETS notation to list the innovations state space models for the 15 univariate ES methods that we compare. We used the *smooth R* package [43] to estimate these models. We limit ourselves to models with additive error. All 15 models have variants with multiplicative error, but these models result in identical equations for computing point forecasts. Given that we use out-of-sample forecast errors to compare methods (rather than in-sample methods that require likelihood values), we gain nothing by includ-

Table 4 ES models that we compare

Trend	Seasonality		
	N	A	M
N	ETS(A,N,N)	ETS(A,N,A)	ETS(A,N,M)
A	ETS(A,A,N)	ETS(A,A,A)	ETS(A,A,M)
A_d	ETS(A, A_d ,N)	ETS(A, A_d ,A)	ETS(A, A_d ,M)
M	ETS(A,M,N)	ETS(A,M,A)	ETS(A,M,M)
M_d	ETS(A, M_d ,N)	ETS(A, M_d ,A)	ETS(A, M_d ,M)

N = none, A = additive, M = multiplicative, d = damped

ing multiplicative error models. Furthermore, multiplicative error models are numerically unstable for data containing zeros [44, Section 2.5.2].

TBATS

De Livera et al [8] proposed the TBATS innovations state-space model, which generalizes the ETS(A,A_d,A) model. TBATS is an acronym for the main features in the model: Trigonometric, Box-Cox transformation, ARMA errors, Trend, and Seasonal components. The seasonal component is modeled as a sum of trigonometric functions, which reduces the number of parameters needed to represent complex seasonal patterns. The Box-Cox transformation includes the square-root transformation recommended by [9]. ARMA errors allow the model to capture more complex dependence within the time series. We used the forecast R package [45] to estimate the TBATS model.

The TBATS updating equations for a time series with a single seasonal pattern with period m are:

$$\begin{aligned} y_{t,r}^{(\omega)} &= \frac{y_{t,r}^\omega - 1}{\omega}, \\ y_{t,r}^{(\omega)} &= l_{t-1,r} + \phi b_{t-1,r} + s_{t-1,r} + d_{t,r}, \\ l_{t,r} &= l_{t-1,r} + \phi b_{t-1,r} + \alpha d_{t,r}, \\ b_{t,r} &= (1 - \phi)b_r + \phi b_{t-1,r} + \beta d_{t,r}, \\ s_{t,r} &= \sum_{j=1}^k s_{j,t,r}, \\ s_{j,t,r} &= s_{j,t-1,r} \cos(\lambda_j t) + s_{j,t-1,r}^* \sin(\lambda_j t) + \gamma_1 d_{t,r}, \\ s_{j,t,r}^* &= -s_{j,t-1,r} \sin(\lambda_j t) + s_{j,t-1,r}^* \cos(\lambda_j t) + \gamma_2 d_{t,r}, \\ \lambda_j &= 2\pi j/m, \\ d_{t,r} &= \sum_{i=1}^p \phi_i d_{t-1,r} + \sum_{i=1}^q \theta_i \epsilon_{t-i,r} + \epsilon_{t,r}, \\ \hat{y}_{t+h,r|t}^{(\omega)} &= l_{t,r} + (\phi + \phi^2 + \dots + \phi^h) b_{t,r} + s_{t-m+h_m^+,r}, \\ \hat{y}_{t+h,r|t} &= \left(\omega \hat{y}_{t+h,r|t}^{(\omega)} + 1 \right)^{1/\omega}, \end{aligned}$$

where $y_{t,r}^{(\omega)}$ is the Box-Cox transformed call volume, $l_{t,r}$ is the level component, $b_{t,r}$ is the trend component, $s_{t,r}$ is the seasonal component, $s_{j,t,r}$ is the stochastic level of the seasonal component, $s_{j,t,r}^*$ is the stochastic growth in the level of the seasonal component, b_r is the long-run trend, $d_{t,r}$ is an ARMA(p, q) process representing the prediction error, and $\epsilon_{t,r}$ is a Gaussian white-noise process with zero mean and constant variance σ^2 . The number of harmonics for the seasonal component is k and λ_j specifies the fre-

quency of the j th harmonic. The parameters to be estimated are $\omega, \phi, \alpha, \beta, \gamma_1, \gamma_2, \phi_i, \theta_i, k, p$, and q .

MEDIC

Setzler et al. [7] provide a forecasting equation for the MEDIC method. In order to express this equation using our notation, we define two sets:

$$\begin{aligned} \tau_1(t) &= \left\{ t - \frac{168}{\delta}, \dots, t - \frac{168}{\delta} \right\} \\ \tau_2(t) &= \{ t' : \text{year}(t) - \bar{y} < \text{year}(t') \leq \text{year}(t), \\ &\quad \text{week}(t') = \text{week}(t), \omega(t') = \omega(t), f(t', \delta) = f(t, \delta) \}, \end{aligned}$$

where $\text{year}(t)$ is the year and $\text{week}(t)$ is the week of the year corresponding to Period t . In words, $\tau_1(t)$ consists of the same period within the week from the preceding $\bar{w}$ weeks and $\tau_2(t)$ consists of the same period within the year, from the preceding $\bar{y}$ years. Using these sets, the MEDIC forecast for period $t + h$ can be calculated as:

$$\hat{y}_{t+h,r|t}^{\delta,\alpha} = \frac{1}{\bar{y} \cdot \bar{w}} \sum_{t' \in \tau_1(t+h)} \sum_{t'' \in \tau_2(t')} y_{t'',r}^{\delta,\alpha}, \text{ for } h \leq 168/\delta. \quad (7)$$

MEDIC requires no parameter estimation, but it requires $\bar{y}$ years of data before the first forecast can be computed and it cannot forecast more than a week into the future. Following [7], we set $\bar{w} = 4$ weeks and $\bar{y} = 5$ years for Calgary and Edmonton, but we reduce $\bar{y}$ to 4 years for Grande Prairie because we have less than 5 years of data for that city.

MV

Ye et al. [9] proposed the multivariate MV time-series forecasting model that incorporates within- and inter-stream dependencies among time series, and, similar to TBATS, includes square-root transformation of call volumes and ARMA error terms. Multivariate forecasting can be beneficial especially when time series exhibit inter-stream lag dependence [9]. We test for such dependence (Appendix D) but do not find evidence for a strong effect. In this section, we introduce the MV model using our notation.

We assume $\delta = 8$ and $\alpha = H$. We use $x_{t,r} = \sqrt{y_{t,r} + 1/4}$, D for the total number of days in the training set, and I for the number of time intervals within a day. For example, for Calgary, at the station catchment area and 8-hr resolution, and for the first training set (consisting of 152 weeks), $m_H = 27$ (the number of ambulance stations), $I = 3$, and $D = 152 \times 7 = 1,064$. We also define the functions, $d(t)$ and

$i(t)$, which specify the day index and time-of-day index of Period t , respectively.

Using our notation, we can write the MV model as

$$\begin{aligned} x_{t,r} &= \sqrt{\lambda_{t,r}} + \epsilon_{t,r}, \quad r = 1, \dots, m, \\ \epsilon_t &= (\epsilon_{t,1}, \epsilon_{t,2}, \dots, \epsilon_{t,m})^\top \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \Sigma), \\ \theta_{t,r} &= \sqrt{\lambda_{t,r}} = u_{d(t),r} f_{i(t),r}^{\omega(t)}, \quad f_{i(t),r}^{\omega(t)} \geq 0, \sum_{i=1}^I f_{i,r}^{\omega(t)} = 1, \\ \mathbf{u}_{d(t)} - \alpha_{\omega(t)} &= \mathbf{A}(\mathbf{u}_{d(t)-1} - \alpha_{\omega(t)-1}) + \mathbf{z}_{d(t)}, \\ \mathbf{z}_{d(t)} &= (z_{d(t),1}, z_{d(t),2}, \dots, z_{d(t),m})^\top \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \Omega), \end{aligned} \quad (8)$$

where $\omega(t)$, as before, denotes the day of the week of Period t . $\mathbf{u}_{d(t)} = (u_{d(t),1}, u_{d(t),2}, \dots, u_{d(t),m})^\top$ is the vector of daily total arrival rates of all regions on the square-root scale, $\alpha_{\omega(t)} = (\alpha_{\omega(t),1}, \alpha_{\omega(t),2}, \dots, \alpha_{\omega(t),m})^\top$ models the day-of-week effect on daily total arrival rate on the square-root scale, and $f_{i(t),r}^{\omega(t)}$ is the intraday rate proportion of Period t for Region r , which depends on the day of week. The weekday indexed by 0 is the same as the weekday indexed by 7. We train Model (8) using the procedure in Section 3.2 of [9].

The h -step-ahead point forecasts are obtained as

$$\begin{aligned} \hat{x}_{DI+h,r} &= \hat{u}_{d(DI+h),r} \hat{f}_{i(DI+h),r}^{\omega(DI+h)}, \\ \hat{\mathbf{u}}_{d(DI+h)} &= \hat{\alpha}_{\omega(DI+h)} + \hat{\mathbf{A}}(\hat{\mathbf{u}}_D - \hat{\alpha}_{\omega(DI)}), \\ \hat{y}_{DI+h,r} &= (\hat{x}_{DI+h,r})^2 - 1/4. \end{aligned}$$

The code used in [9] to implement MV is not publicly available. We implemented the model using the R statistical language (see Appendix E for details and a link to our code). The numerical results in [9] are for two streams, whereas our results are for up to 27 streams (the number of station catchment areas in Calgary). Training the model for Calgary required the estimation of 30,969 parameters.

5.2 Hierarchical forecasting

Hierarchical forecasting frameworks provide a way to obtain reconciled forecasts for multiple related time series, whose relationships can be represented in a hierarchical structure [22]. Figure 6(a), for example, shows the temporal hierarchy for our time series at different temporal resolutions, corresponding to different time intervals. Each node in Fig. 6(a) denotes a time series for a time interval specified by its start and end time (not just duration). Intuitively, we expect a time-series forecast to equal the sum of the forecasts for the time series immediately below it in the hierarchy. For example, we expect forecasts for (8, 16] to equal the sum of the forecasts for (8, 12] and (12, 16]. This might not happen, however, if the time series are forecast independently. Hierarchical forecasting involves changing the original forecasts in a post hoc way, such that the *reconciliation error* is minimized. Reconciliation error measures the amount by which time-series forecasts differ from the expected value of bottom-up forecasts based on the most disaggregated time series. Past studies have treated 1-dimensional hierarchies, in which time series are related along a single dimension. We extend this work to 2-dimensional hierarchies, in which time series are related along both a time and a space dimension, as illustrated in Fig. 6. For example, the Calgary time series at the city-level spatial resolution and 24-hour temporal resolution is connected in the 2-dimensional hierarchy to $3 \times 5 = 15$ time series: the three time series for (0, 8], (8, 16], and (16, 24]; and the five time series for Regions 1 to 5.

We define vectors of time-series forecasts, ordered according to a predefined sequence, to make it easier to access a time series at a particular location in the hierarchy. Let $\hat{\mathbf{y}}_{t+h}^{\delta,\alpha}$ be the vector of forecasts for a δ -level temporal and α -level spatial resolution, made at the end of Period t , for the h^{th} 24-hour time period:

$$\hat{\mathbf{y}}_{t+h}^{\delta,\alpha} = \left[\hat{y}_{t+24(h-1)/\delta+1,1}^{\delta,\alpha}, \dots, \hat{y}_{t+24(h)/\delta,1}^{\delta,\alpha}, \dots, \hat{y}_{t+24(h-1)/\delta+1,m_\alpha}^{\delta,\alpha}, \dots, \hat{y}_{t+24(h)/\delta,m_\alpha}^{\delta,\alpha} \right].$$

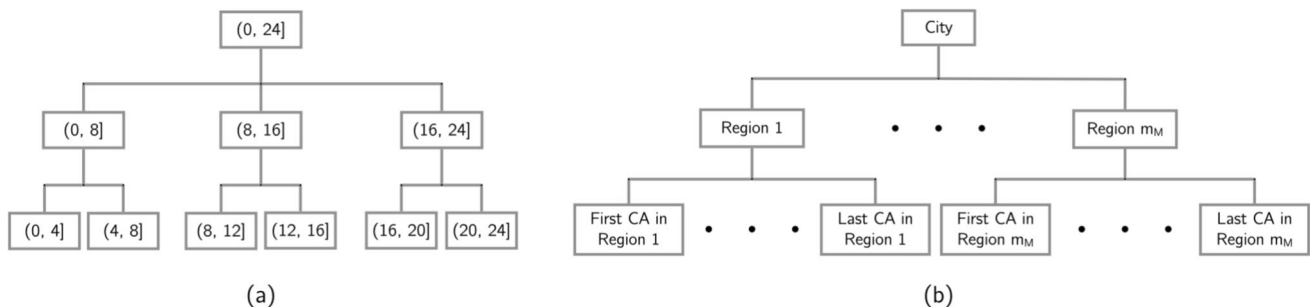


Fig. 6 The (a) temporal and (b) spatial dimensions of the hierarchical structure of our time series. CA = Catchment area

We collect all of the original forecasts, before reconciliation, in $\hat{\mathbf{y}}_{t+h}$, the vector of *base forecasts*:

$$\hat{\mathbf{y}}_{t+h} = [\hat{y}_{t+h}^{24,L}, \hat{y}_{t+h}^{24,M}, \hat{y}_{t+h}^{24,H}, \hat{y}_{t+h}^{8,L}, \hat{y}_{t+h}^{8,M}, \hat{y}_{t+h}^{8,H}, \hat{y}_{t+h}^{4,L}, \hat{y}_{t+h}^{4,M}, \hat{y}_{t+h}^{4,H}].$$

The vector of base forecasts for Calgary, for example, has 330 elements.

We obtain bottom-up forecasts for all time series in the hierarchy by summing the forecasts of the most disaggregated time series ($\delta = 4$ and $\alpha = H$), using the *summing matrix* $\mathbf{S}$:

$$\mathbf{S} \hat{\mathbf{y}}_{t+h}^{4,H\tau},$$

where $\mathbf{S}_{ij} = 1$ if the time interval and spatial area of the j^{th} entry of $\hat{\mathbf{y}}_{t+h}^{4,H}$ are within the time interval and spatial area of the i^{th} entry of $\hat{\mathbf{y}}_{t+h}$, and $\mathbf{S}_{ij} = 0$ otherwise. For Calgary, $\mathbf{S}$ has 330 rows and 162 columns. We decompose the base forecasts as

$$\hat{\mathbf{y}}_{t+h}^{\tau} = \mathbf{S} \boldsymbol{\beta}_h + \boldsymbol{\epsilon}_h,$$

where $\boldsymbol{\epsilon}_h$ is the reconciliation error, and $\boldsymbol{\beta}_h$ is the unknown conditional mean of the future values of the most disaggregated time series. Assuming that $\boldsymbol{\epsilon}_h$ has zero mean and a known covariance matrix $\boldsymbol{\Sigma}$, we can compute reconciled forecasts, $\tilde{\mathbf{y}}_{t+h}$, using a generalized least-squares estimate of $\boldsymbol{\beta}_h$ [46]:

$$\tilde{\mathbf{y}}_{t+h}^{\tau} = \mathbf{S}(\mathbf{S}^{\tau} \boldsymbol{\Sigma}^{-1} \mathbf{S})^{-1} \mathbf{S}^{\tau} \boldsymbol{\Sigma}^{-1} \hat{\mathbf{y}}_{t+h}^{\tau}. \quad (9)$$

Following [46], we approximate $\boldsymbol{\Sigma}$ by a diagonal matrix, $\hat{\boldsymbol{\Sigma}}$:

$$\begin{aligned} \hat{\boldsymbol{\Sigma}} = \text{diag}(\hat{\sigma}_1^{24,L}, \hat{\sigma}_1^{24,M}, \dots, \hat{\sigma}_{m_M}^{24,M}, \hat{\sigma}_1^{24,H}, \dots, \hat{\sigma}_{m_H}^{24,H}, \\ \hat{\sigma}_1^{8,L} \mathbf{e}_3, \hat{\sigma}_1^{8,M} \mathbf{e}_3, \dots, \hat{\sigma}_{m_M}^{8,M} \mathbf{e}_3, \hat{\sigma}_1^{8,H} \mathbf{e}_3, \dots, \hat{\sigma}_{m_H}^{8,H} \mathbf{e}_3, \\ \hat{\sigma}_1^{4,L} \mathbf{e}_6, \hat{\sigma}_1^{4,M} \mathbf{e}_6, \dots, \hat{\sigma}_{m_M}^{4,M} \mathbf{e}_6, \hat{\sigma}_1^{4,H} \mathbf{e}_6, \dots, \hat{\sigma}_{m_H}^{4,H} \mathbf{e}_6), \end{aligned}$$

where $\hat{\sigma}_i^{\delta,\alpha}$ is the in-sample, one-step-ahead error variance of the time series for Region i at temporal resolution δ and spatial resolution α , and $\mathbf{e}_d$ is a d -dimensional row vector of ones.

6 Performance measures

In this section, we define performance measures and discuss how we aggregate or summarize performance across forecast horizons and spatio-temporal resolutions. We use out-of-sample forecast accuracy rather than in-sample measures like AIC, because we have a sufficiently large amount of data [40].

6.1 Error measures

We use the mean absolute error (MAE) and the mean absolute scaled error (MASE) as measures of forecast performance. Given a sequence of out-of-sample errors, $e_1, e_2, \dots, e_h$, the MAE is:

$$\text{MAE} = \frac{\sum_{i=1}^h |e_i|}{h}.$$

MAPE is a common scale-free alternative to MAE but MAPE cannot be computed if the data contains zeros. MASE addresses this issue [47]. To calculate MASE we first calculate the average in-sample error of a naive method that forecasts y_t as y_{t-m} , where m is the length of the seasonal pattern:

average in-sample error of the

naive method $= e_{\text{naive}}$

$$= \frac{1}{\tau - m} \sum_{j=m+1}^{\tau} |y_j - y_{j-m}|,$$

where τ is the number of observations in the training set. We scale each out-of-sample error, $e_1, e_2, \dots, e_h$, by dividing it with e_{naive} : $q_i = e_i / e_{\text{naive}}$. We calculate MASE as:

$$\text{MASE} = \frac{\sum_{i=1}^h |q_i|}{h}.$$

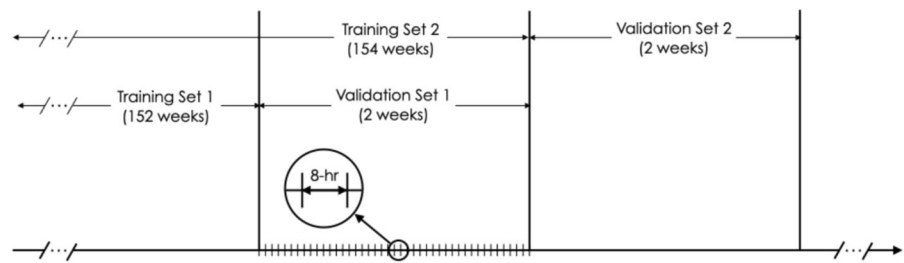
The way in which MASE is defined permits us to interpret $\text{MASE} < 1$ as indicating a method whose out-of-sample performance is better than the in-sample performance of the naive method. The naive method in [47] uses y_{t-1} as a forecast for y_t . Our naive method, which uses y_{t-m} as a forecast for y_t , is more sophisticated, in that it accounts implicitly for weekly seasonality.

6.2 Out-of-sample rolling forecast

We compute out-of-sample rolling forecasts as illustrated in Fig. 7. Our first training set consists of the first 152 weeks of data (108 weeks for Grande Prairie), with the following 2 weeks serving as the validation set. We extend the training set by 2 weeks and shift the validation set forward by 2 weeks and repeat that process until we run out of data. This process results in 50-fold out-of-sample validation. Following [48], we take the following steps to compare the performance of different methods:

1. For each city, method, and Fold, f , calculate MAE_f and MASE_f using the out-of-sample forecast errors for all combinations of station catchment area and forecast horizon.

Fig. 7 Out-of-sample validation, illustrated for an 8-hour temporal resolution



2. Compute averages across folds:

$$\overline{\text{MAE}} = \frac{1}{F} \sum_{f=1}^{50} \text{MAE}_f, \quad (10)$$

$$\overline{\text{MASE}} = \frac{1}{F} \sum_{f=1}^{50} \text{MASE}_f, \quad (11)$$

where F is the number of folds. Averaging across folds leads to lower estimation variance, while using out-of-sample error measures allows us to compare the generalization error of different methods.

6.3 Identifying methods that are within $p\%$ of the best method

The approach in Section 6.2 averages, first, across station catchment areas and forecast horizons and, second, across folds. We now describe an alternative approach, which allows us to identify methods that perform within $p\%$ of the best method, and to compare methods separately for each station catchment area and forecast horizon. We use the scale-free MASE measure for this approach.

For each city, Method m , and Station Catchment Area s , we compute $\text{MASE}_{m,s}$ using the out-of-sample forecast errors at all forecast horizons, and average across folds to obtain $\overline{\text{MASE}}_{m,s}$. Next, we compute the proportion of station

catchment areas for which Method m is within p percent of the lowest MASE, that is:

$$W_{p,m}^{\text{SCA}} = \frac{1}{S} \sum_s \mathbb{1} \left\{ \overline{\text{MASE}}_{m,s} \leq (1 + p/100) \min_m \overline{\text{MASE}}_{m,s} \right\}, \quad (12)$$

where S is the number of station catchment areas and $\mathbb{1}$ is the indicator function. Similarly, for each city, Method m , and Horizon h , we compute $\text{MASE}_{m,h}$ using the out-of-sample forecast errors for all station catchment areas, and average across folds to obtain $\overline{\text{MASE}}_{m,h}$, and use it to compute the proportion of forecast horizons for which Method m is within p percent of the lowest MASE, that is:

$$W_{p,m}^{\text{HOR}} = \frac{1}{H} \sum_h \mathbb{1} \left\{ \overline{\text{MASE}}_{m,h} \leq (1 + p/100) \min_m \overline{\text{MASE}}_{m,h} \right\}, \quad (13)$$

where H is the number of horizons.

6.4 Comparison of methods with different spatial and temporal resolutions

We use 9 spatio-temporal resolutions for estimation, as shown in the first two columns of Table 5. In comparing methods, however, we measure performance after converting forecasts to a common spatio-temporal resolution, namely,

Table 5 Spatio-temporal resolution levels (comparison level in bold)

Spatial resolution for estimation	Temporal resolution for estimation	Spatial forecast	Temporal forecast
Catchment area	4-hr	Same level	Bottom-up
Catchment area	8-hr	Same level	Same level
Catchment area	24-hr	Same level	Top-down
Intermediate region	4-hr	Top-down	Bottom-up
Intermediate region	8-hr	Top-down	Same level
Intermediate region	24-hr	Top-down	Top-down
City	4-hr	Top-down	Bottom-up
City	8-hr	Top-down	Same level
City	24-hr	Top-down	Top-down

catchment areas and 8-hour time periods. We combine and divide forecasts independently in space and in time, in order to convert to the comparison level, as previewed in the last two columns of Table 5 and outlined in the next two subsections.

6.5 Converting forecasts from higher to lower resolution

Bottom-up forecasting involves the combination of forecasts for multiple regions or multiple time intervals to obtain a forecast for a larger region or a longer time interval. We do this simply by adding forecasts. For example, we add two 4-hour forecasts to obtain an 8-hour forecast:

$$\hat{y}_{t,r}^{8,\alpha} = \sum_{\tau: f(\tau,4) \subset f(t,8)} \hat{y}_{\tau,r}^{4,\alpha}$$

6.6 Converting forecasts from lower to higher resolution

Top-down forecasting involves the decomposition of a forecast made at a low resolution to multiple forecasts at a higher resolution. We begin by detailing how we decompose forecasts in time. Our general approach is to estimate weights using the smallest training set (Fold 1) and use those weights to decompose forecasts.

We use the following equations to divide a 24-hour forecast into three 8-hour forecasts:

$$\hat{y}_{t,r}^{8,\alpha} = \omega_1(g(t)) \hat{y}_{\tau,r}^{24,\alpha} \text{ for all } t : f(t, 8) \subset f(\tau, 24),$$

$$\omega_1(\rho) = \frac{N_\rho}{\sum_{j=1}^{21} N_j} \text{ for all } \rho = 1, \dots, 21,$$

where $\omega_1(\rho)$ is an estimate of the percentage of the weekly call volume that occurs in Interval ρ , and N_ρ is the number of calls in the Fold 1 training set that occurred in Interval ρ .

We use the following equations to divide a forecast with a spatial resolution of α to forecasts at a higher spatial resolution of α' :

$$\hat{y}_{t,s}^{\delta,\alpha'} = \omega_2(r, s) \hat{y}_{t,r}^{\delta,\alpha}, \text{ for all } s \in R(\alpha'), r \in R(\alpha), s \text{ is part of } r,$$

$$\omega_2(r, s) = \frac{M_s}{M_r},$$

where $\omega_2(r, s)$ is an estimate of the percentage of the call volume from Region $r \in R(\alpha)$ that occurs in Region $s \in R(\alpha')$, and M_s and M_r are the numbers of calls in the Fold 1 training set that occurred in Regions s and r , respectively.

7 Top-down vs. hierarchical forecasting: A simulation study

To motivate the simulation study that we discuss in this section, suppose that call-volume forecasts are desired for every hour of the week for each of the 4 intermediate regions in Edmonton. Our top-down approach forecasts city-wide call volume for each hour, using ETS(A,N,A), and then disaggregates to intermediate regions. Our bottom-up approach uses reconciled base forecasts for each intermediate region and each 1-hour period, using a set of base forecasts computed using ETS(A,N,A).

The top-down approach has the benefits of simplicity of implementation and fewer parameters to estimate. The potential disadvantage of the top-down approach is that the disaggregation procedure implicitly assumes that the seasonal patterns and the spatial distribution combine in a multiplicative fashion, similar to the notion of statistical independence. Contrary to this implicit assumption, Fig. 5 demonstrates that the weekly seasonal patterns for the 5 intermediate regions in Calgary are different. In particular, the pattern for Region 1 (the central business district) is different from the rest of the city. The same is true for Edmonton. Based on this exploratory analysis, one might expect a top-down approach to perform poorly.

We use simulated data to explore factors that might cause a top-down approach to perform poorly. We use Edmonton data for intermediate regions and hourly call volumes to develop a base scenario. We perturb the base scenario to develop 3 scenarios corresponding to each of 3 factors: (1) the amplitude of Region 1's weekly seasonality, (2) a phase shift of Region 1's weekly seasonality, and (3) the city-level call volume. Table 6 summarizes the $1 + 3 \times 3 = 10$ scenarios. We hypothesize that a top-down approach will perform worse as the amplitude and phase shift of Region 1 (compared to the rest of the city) increase, and will perform better as the city-wide call volume decreases.

We simulate the call volume for Region i and Hour t under Scenario s as Poisson-distributed with mean $\rho_{i,t}^s n^s$, where

Table 6 Summary of scenarios

Scenarios	Specification
B: Base	Edmonton, intermediate regions, 1-hour periods
A1–A3: Vary Region 1 amplitude	$\alpha^s = 1.2, 1.41, 1.61$
P1–P3: Vary Region 1 phase shift	$\delta^s = 3, 6, 9$ hours
V1–V3: Vary city-wide call volume	Reduce by a factor of 5, 10, 20

$i = 1, 2, 3, 4$ and $t = 0, 1, \dots, 43, 679$ (corresponding to a total of five years); $\rho_{i,t}^s$ denotes the proportion of the city's weekly call volume for Region i and Hour t under Scenario s , and n^s denotes the city-wide average weekly call volume under Scenario s .

For the base scenario, based on Edmonton data for 2016, we set $n^B = 1,717$ and we set $\rho_{i,t}^B$ equal to the average proportion of the weekly city-wide call volume that occurred in Hour $\text{mod}(t, 168)$ of the week and in Region i . We number the hours of the week from 0 to 167, with Hour 0 corresponding to 12 to 1 am on Monday.

Scenarios A1–A3 differ from B only for the Region 1 proportions. We define $\bar{\rho}_1 = \sum_{t=0}^{167} \rho_{1,t}^B / 168$ as the average proportion per hour for Region 1, and set $\rho_{1,t}^s = \bar{\rho}_1 + \alpha^s(\rho_{1,t}^B - \bar{\rho}_1)$, for $s = A1, A2, A3$, and all t . The parameter α^s controls how much the amplitude is increased. We use $\alpha^{A1} = 1.2$, $\alpha^{A2} = 1.41$, and $\alpha^{A3} = 1.61$ (the maximum possible before some proportions become negative). Scenarios P1–P3 differ from B only for the Region 1 proportions, which we set to $\rho_{1,t}^s = \rho_{1, \text{mod}(t-\delta^s, 168)}^B$. The parameter δ^s controls the magnitude of the phase shift, and is set to $\delta^{P1} = 3$, $\delta^{P2} = 6$, $\delta^{P3} = 9$ hours. Scenarios V1–V3 differ from B only for the city-wide average weekly call volume, which we set to $n^{V1} = n^B/5 = 343$, $n^{V2} = n^B/10 = 172$, and $n^{V3} = n^B/20 = 86$.

We simulate 10 five-year replications for each scenario. We use common random numbers to reduce variance for comparisons across scenarios. For each replication, we train a bottom-up and a top-down ES method with additive seasonality and no trend. For each scenario, replication, and region, we calculate $\overline{\text{MAE}}$ using the out-of-sample rolling horizon approach described in Section 6. For each scenario and region, we average the $\overline{\text{MAE}}$ values across replications to obtain $\overline{\overline{\text{MAE}}}$. Finally, for each scenario, we average the $\overline{\overline{\text{MAE}}}$ values across regions to obtain $\overline{\overline{\overline{\text{MAE}}}}$.

Table 7 $\overline{\overline{\overline{\text{MAE}}}}$ values for the bottom-up and top-down approaches for all scenarios, and the percent increase in $\overline{\overline{\overline{\text{MAE}}}}$ if using a top-down approach

Scenario	Bottom-up	Top-down	Cost of using top-down
B	1.262	1.263	0.12%
A1	1.259	1.265	0.46%
A2	1.257	1.269	0.99%
A3	1.253	1.277	1.88%
P1	1.261	1.290	2.26%
P2	1.261	1.323	4.91%
P3	1.261	1.341	6.36%
V1	0.384	0.385	0.21%
V2	0.220	0.220	0.18%
V3	0.585	0.588	0.44%

Table 8 $\overline{\overline{\overline{\text{MAE}}}}$ values for the bottom-up and top-down approaches for all scenarios, and the percent increase in $\overline{\overline{\overline{\text{MAE}}}}$ if using a top-down approach

Scenario	Bottom-up	Top-down	Cost of using top-down
B	1.438	1.424	−0.99%
A1	1.464	1.465	0.06%
A2	1.490	1.505	1.02%
A3	1.525	1.556	2.02%
P1	1.395	1.410	1.04%
P2	1.281	1.382	7.92%
P3	1.167	1.372	17.60%
V1	0.465	0.461	−0.86%
V2	0.263	0.268	1.82%
V3	0.676	0.677	0.06%

Results are for Region 1, for Saturdays from 12 pm to midnight

7.1 Results

Table 7 shows $\overline{\overline{\overline{\text{MAE}}}}$ results for the top-down and bottom-up approaches for all scenarios. Table 8 shows separate $\overline{\overline{\overline{\text{MAE}}}}$ results for forecasts of Region 1's call volume on Saturday from 12 pm to midnight—the time period during which Region 1's seasonal pattern differs the most from the city-wide pattern. We observe that, as expected, the “cost” of using the top-down approach increases as the amplitude and phase shift of the Region 1 seasonal pattern increase. The increase with respect to the phase shift is more pronounced for Region 1 on Saturdays but this is not the case for the amplitude. Somewhat surprisingly, we see a negligible effect of total average call volume.

8 Empirical results and performance of forecasting methods

In this section we compare the out-of-sample performance of the various methods. First, we compare methods with differ-

Table 9 Performance of methods with different components for Calgary

Method	Trend	Seasonality	$\overline{\text{MAE}}$	$\overline{\text{MASE}}$
ETS(A,N,N)	None	None	1.63	89.4%
ETS(A,A,N)	Additive	None	1.64	89.8%
ETS(A,N,A)	None	Additive	1.45	79.1%
ETS(A,A,A)	Additive	Additive	1.46	79.5%
TBATS	Additive	Additive	1.45	79.1%
MV	None	Additive	1.47	80.0%
MEDIC	None	None	1.47	80.1%

The mean and standard deviation of the test-set call volume are 3.29 and 2.74, respectively

Table 10 Performance of methods with different components for Edmonton

Method	Trend	Seasonality	MAE	MASE
ETS(A,N,N)	None	None	2.10	89.3%
ETS(A,A,N)	Additive	None	2.10	89.6%
ETS(A,N,A)	None	Additive	1.81	76.4%
ETS(A,A,A)	Additive	Additive	1.82	76.9%
TBATS	Additive	Additive	1.81	76.5%
MV	None	Additive	1.86	78.3%
MEDIC	None	Additive	1.85	78.0%

The mean and standard deviation of the test-set call volume are 5.06 and 3.57, respectively

ent time-series components. Second, we compare methods that differ in the spatio-temporal resolution used for estimation, with the number and type of time-series components fixed to allow a fair comparison. Third, we compare hierarchical and non-hierarchical methods.

8.1 Comparison between methods with different components

Methods with a multiplicative component cannot be estimated if the data contains zeros. We estimated all models in Table 4 using a spatio-temporal resolution for which the data had no zeros. We found that methods with multiplicative seasonality or multiplicative or damped trend provided no benefit over methods with only additive components (see Appendix B). Therefore, in what follows, we limit our discussion to methods with additive components, which can handle zeros.

Tables 9–11 show a comparison of methods, in terms of $\overline{\text{MAE}}$ and $\overline{\text{MASE}}$, computed as shown in Eqs. 10–11, for Calgary, Edmonton, and Grande Prairie, respectively. We adjusted the number of folds F for computing the performance measures to include only folds available with the MEDIC method.

The spatio-temporal resolution was fixed at $\delta = 8$ and $\alpha = H$. We show the mean and the standard deviation of the

Table 11 Performance of methods with different components for Grande Prairie

Method	Trend	Seasonality	MAE	MASE
ETS(A,N,N)	None	None	1.83	79.8%
ETS(A,A,N)	Additive	None	1.83	80.0%
ETS(A,N,A)	None	Additive	1.69	72.8%
ETS(A,A,A)	Additive	Additive	1.72	74.1%
TBATS	Additive	Additive	1.70	72.9%
MEDIC	Additive	Additive	1.71	73.7%

The mean and standard deviation of the test-set call volume are 4.23 and 2.35, respectively

call volumes in the table captions, to facilitate interpretation of the MAE values.

All three tables indicate that, for the ETS methods, including a seasonal component improves performance but the inclusion of a trend component has minimal impact—despite the fact that the time series appear to have a strong trend component (see Table 3). All methods outperform the naive approach, as shown by $\overline{\text{MASE}}$ values below 1.

The ETS(A,N,A) method is either best or tied for best, for all 3 cities, and TBATS is never far behind. The MV and MEDIC methods perform better than ETS methods with no seasonality component but worse than TBATS and ETS methods with a seasonality component.

Table 12 shows $W_{p,m}^{\text{SCA}}$ and $W_{p,m}^{\text{HOR}}$, computed using Eqs. 12–13, for $p = 1\%$ and 5% , for Calgary. The results in Table 12 complement and are largely consistent with those in Table 9. The finding that including seasonality is important but including trend is not holds, not only on average for all of Calgary, but also for most individual station catchment areas and for most individual forecast horizons. For example, the MASEs for ETS(A,N,A), ETS(A,A,A), and TBATS are within 5% of the best-performing method for all station catchment areas and for all forecast horizons. We observe similar patterns for Edmonton and Grande Prairie (detailed results not shown).

Figure 8 shows the the proportion of station catchment areas in Calgary for which each method was within 5% of the lowest MASE for different forecast horizons. The top (bottom) panel shows methods with (without) a seasonality component. The methods with no seasonal component are almost always worse than the methods with a seasonality component. The number of horizons, out of 42, for which each method performed best (or is tied for best) is 15 for ETS(ANA), 14 for TBATS, 13 for MV, 9 for ETS(AAA), 5 for ETS(AAN), and 1 for ETS(ANN). We see pronounced and systematic seasonality in the performance of the methods that do not include a seasonal component. The results for Edmonton are similar, with ETS(ANA), TBATS, MV, ETS(AAA), ETS(AAN), and ETS(ANN) performing best for 17, 20, 9, 8, 1, and 0 of the 42 horizons, respectively.

8.2 Comparison between methods with different resolutions

In order to compare different resolutions, we fix the method to be TBATS (one of the top two performers, based on Section 8.1). Table 13 compares forecasts obtained using different temporal and spatial resolutions for estimation. Forecasts are brought to the same resolution (8-hour and catchment area) through bottom-up and top-down approaches (see Section 6).

The simulation results in Section 7 suggest that a top-down approach may perform poorly. The results on real-world data

Table 12 The proportion of station catchment areas (SCAs) and forecast horizons for which each method has a performance within 1% and 5% of the MASE of the best method, for Calgary

Method	Trend	Seasonality	Across SCAs		Across horizons	
			1%	5%	1%	5%
ETS(A,N,N)	None	None	0.0%	3.7%	11.9%	50.0%
ETS(A,A,N)	Additive	None	0.0%	7.4%	11.9%	50.0%
ETS(A,N,A)	None	Additive	77.8%	100.0%	78.6%	100.0%
ETS(A,A,A)	Additive	Additive	63.0%	100.0%	64.3%	100.0%
TBATS	Additive	Additive	85.2%	100.0%	78.6%	100.0%
MV	None	Additive	51.9%	92.6%	38.1%	95.2%
MEDIC	None	Additive	33.3%	96.3%	28.6%	100.0%

(Table 13), however, indicate that there is little difference between the various top-down, bottom-up, and same-level methods. This means that, methods with a 24-hour temporal resolution, and thus no daily seasonal component, perform as well as methods with a seasonal component. The reason for this is that by using historical proportions to divide the 24-hour forecasts across 8-hour intervals we implicitly take the daily seasonality of call volumes into account. The good performance of such a top-down approach is in accordance with past research [21].

Inspired by the good performance of the top-down approach, we also tried a weekly temporal resolution, at both the station catchment area and city spatial resolutions, as shown in the last two rows of Table 13. We see that this simple approach performs quite well—and in one case (Calgary, 1-week, Station) it is the best-performing approach, according to the $\overline{\text{MASE}}$ measure.

8.3 Comparison between reconciled and base forecasts

Table 14 compares the base and reconciled forecasts for all cities and all resolutions. The hierarchical approach provides reconciled forecasts for all resolutions simultaneously. We

found that the reconciled forecasts differed from the base forecasts in all cases, but the differences were smallest for the highest resolutions. As a result, at the 8-hour and station catchment area resolution—the resolution appropriate for planning—the base and reconciled forecasts have almost identical accuracy. The reconciled forecasts, however, perform better than the base forecasts at the city-level spatial resolution, for Calgary and Edmonton.

9 Interpretation of the top ES method and two of the benchmark methods

In this section, we compare how forecasts are computed using the best-performing ES method (ETS(A,N,A)) as well as two of the benchmark methods (TBATS and MV). We fix the training set to be the first training set for Calgary's Region 1 time series, at the 8-hr temporal resolution. This time series has no zeros and, thus, a Box-Cox transformation (a component of TBATS) is possible. We omit the r subscript for ETS(A,N,A) and TBATS but keep it for MV, because with this method information for regions $r = 2, 3, 4$, and 5 is needed to compute the forecast for $r = 1$.

Fig. 8 The proportion of station catchment areas for which each method has a performance within 5% of the best method at different forecast horizons for Calgary

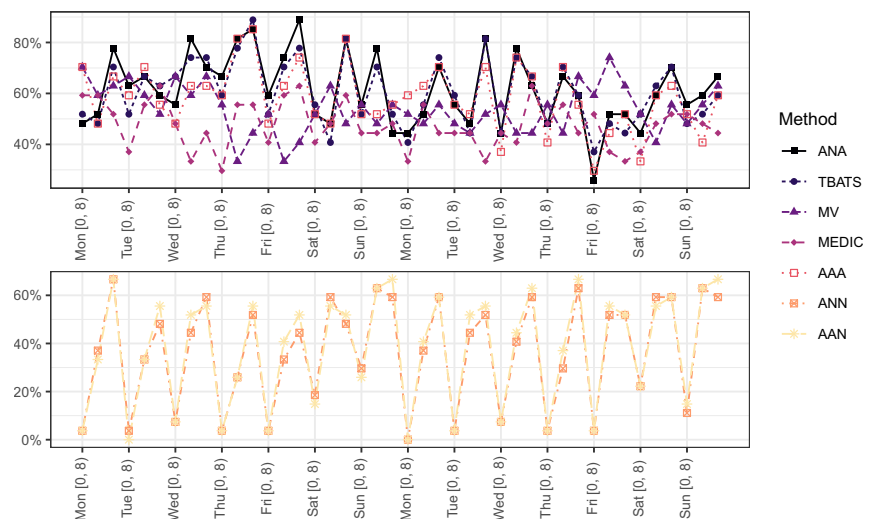


Table 13 Comparison of spatio-temporal resolutions used for estimation

δ	α	Calgary		Edmonton		Grande Prairie	
		MAE	MASE	MAE	MASE	MAE	MASE
4-hour	Station	1.44	78.5%	1.82	76.3%		
4-hour	Intermediate	1.44	78.4%	1.82	76.8%		
4-hour	City	1.44	78.3%	1.83	76.8%	1.73	74.7%
8-hour	Station	1.43	77.9%	1.80	75.7%		
8-hour	Intermediate	1.43	77.9%	1.81	76.0%		
8-hour	City	1.44	78.1%	1.82	76.2%	1.72	74.3%
24-hour	Station	1.44	78.0%	1.81	75.7%		
24-hour	Intermediate	1.44	78.1%	1.82	76.2%		
24-hour	City	1.44	78.2%	1.82	76.4%	1.72	74.1%
1-week	Station	1.43	77.7%	1.80	75.7%		
1-week	City	1.44	78.2%	1.82	76.4%	1.72	74.3%

Method is fixed to TBATS

The updating equations for the trained ETS(A,N,A) method are:

$$\begin{aligned}\hat{y}_t &= l_{t-1} + s_{t-21}, \\ l_t &= 0.054(y_t - s_{t-21}) + 0.946(l_{t-1}), \\ s_t &= s_{t-21}.\end{aligned}\quad (14)$$

The smoothing parameter for the seasonal component is zero, meaning that the seasonal indices are not updated once initialized. Table 15 illustrates how the ETS(A,N,A) forecast calculations could be explained to a non-technical audience in an easily interpretable format.

The updating equations for the trained TBATS method are:

$$\begin{aligned}y_t^{(0.239)} &= \frac{y_t^{0.239} - 1}{0.239}, \\ \hat{y}_t^{(0.239)} &= l_{t-1} + 0.937b_{t-1} + s_{t-1}, \\ l_t &= l_{t-1} + 0.937b_{t-1}, \\ b_t &= 0.17b + 0.937b_{t-1}, \\ s_t &= \sum_{j=1}^9 s_{j,t}, \\ s_{j,t} &= s_{j,t-1} \cos(\lambda_j t) + s_{j,t-1}^* \sin(\lambda_j t), \\ s_{j,t}^* &= -s_{j,t-1} \sin(\lambda_j t) + s_{j,t-1}^* \cos(\lambda_j t), \\ \lambda_j &= 2\pi j/21.\end{aligned}\quad (15)$$

TBATS uses a Box-Cox transformation with $\omega = 0.239$, whereas a square-root transformation (proposed in [49] to normalize the distribution and stabilize its variance) corresponds to $\omega = 0.5$. No ARMA error terms are included. Trend is included, and is decomposed into a short-run (b_t)

Table 14 Comparison of MAE of reconciled and base forecasts

δ	α	Calgary		Edmonton		Grande Prairie	
		Base	Recon.	Base	Recon.	Base	Recon.
4-hour	Station	1.00	1.00	1.25	1.25		
4-hour	Intermediate	2.46	2.45	2.65	2.64		
4-hour	City	6.04	5.99	5.75	5.65	1.20	1.20
8-hour	Station	1.43	1.43	1.81	1.81		
8-hour	Intermediate	3.57	3.56	3.87	3.86		
8-hour	City	9.10	9.04	8.82	8.63	1.72	1.72
24-hour	Station	2.55	2.56	3.24	3.25		
24-hour	Intermediate	6.62	6.58	7.15	7.17		
24-hour	City	18.17	17.87	17.45	16.85	3.02	3.02

Method is fixed to ETS(A,N,A)

Table 15 Illustration of ETS(A,N,A) forecast calculations, assuming that the current time is the beginning of 2 March 2015, the current level is $l_t = 83.82$, and the seasonal differences s_t are as shown

Date	Period start	Period end	$L = \text{Level}$	$SD = \text{Seasonal differences}$	Forecast (calls) $F = L + SD$
2 March 2015	12 AM	8 AM	83.82	-30.32	53.50
	8 AM	4 PM	83.82	18.35	102.17
	4 PM	12 AM	83.82	9.05	92.87
3 March 2015	12 AM	8 AM	83.82	-33.74	50.086
	8 AM	4 PM	83.82	19.44	103.26
	4 PM	12 AM	83.82	9.80	93.62
4 March 2015	12 AM	8 AM	83.82	-34.32	49.50
	8 AM	4 PM	83.82	19.55	103.37

and long-run (b) trend. Seasonality is modeled as the sum of 9 harmonics.

The updating equations for the trained MV method are:

$$\hat{x}_{t,1} = \sqrt{\hat{y}_{t,1} + 1/4} = \hat{u}_{d(t),1} \hat{f}_{i(t),1}^{\omega(t)}, \quad (16)$$

$$\hat{u}_{d(t),1} - \hat{\alpha}_1^{\omega(t)} = [0.27, 0.11, 0.02, 0.08, 0.08] \begin{bmatrix} \hat{u}_{d(t)-1,1} - \hat{\alpha}_1^{\omega(t)-1} \\ \hat{u}_{d(t)-1,2} - \hat{\alpha}_2^{\omega(t)-1} \\ \hat{u}_{d(t)-1,3} - \hat{\alpha}_3^{\omega(t)-1} \\ \hat{u}_{d(t)-1,4} - \hat{\alpha}_4^{\omega(t)-1} \\ \hat{u}_{d(t)-1,5} - \hat{\alpha}_5^{\omega(t)-1} \end{bmatrix},$$

with

$$\begin{aligned} \hat{\mathbf{f}}_1^T &= (\hat{f}_{1,1}^1, \hat{f}_{2,1}^1, \hat{f}_{3,1}^1, \hat{f}_{1,1}^2, \dots, \hat{f}_{3,1}^7)^T \\ &= (0.33, 0.32, 0.35, 0.28, 0.36, 0.36, 0.26, 0.37, 0.37, 0.26, 0.37, \\ &\quad 0.36, 0.27, 0.36, 0.37, 0.27, 0.35, 0.38, 0.31, 0.31, 0.38)^T, \end{aligned}$$

$$\begin{aligned} \hat{\alpha}^T &= (\hat{\alpha}_1^1, \hat{\alpha}_1^2, \dots, \hat{\alpha}_1^7, \hat{\alpha}_2^1, \dots, \hat{\alpha}_5^7)^T \\ &= (11.30, 11.01, 10.96, 11.05, 11.04, 11.29, 11.44, 09.47, 09.36, 09.44, 09.31, \\ &\quad 09.33, 09.36, 09.39, 11.15, 11.05, 10.99, 10.88, 11.05, 11.10, 11.21, 13.56, \\ &\quad 13.51, 13.38, 13.30, 13.57, 13.45, 13.52, 15.13, 15.31, 15.20, 15.16, 15.07, \\ &\quad 15.31, 15.17)^T, \end{aligned}$$

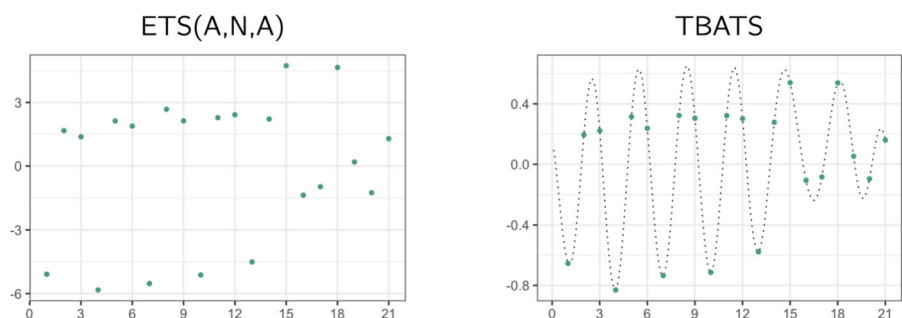
and $\hat{\mathbf{u}}$ a 1751-length vector containing estimates of daily total arrival rates (on the square-root scale). Model (16) first

estimates Region 1's seasonal-adjusted daily call volume—using information from all regions within the city—and then, in a top-down manner, decomposes the daily forecast to 8-hr period forecasts, using weights ($\hat{f}_{1,i}^{\omega}$) that depend on both the region and weekday.

Comparing (14) with Eqs. 15 and 16, we see that the relationship between the components and the way in which they contribute to the final forecast is much clearer for ETS(A,N,A) than for TBATS and MV. It is much easier to understand how forecasts are calculated by ETS(A,N,A) than for TBATS and MV. This is especially true for forecast users without a strong technical background. Furthermore, ETS(A,N,A) estimates only 24 parameters, as opposed to the 1,812 parameters needed to train MV, not including the parameters for other regions and covariance matrices for the error terms. A high number of parameters carries the risk of overfitting. Increasing the size of the training set, a common remedy for overfitting, might not be effective for MV because the number of parameters increases with the size of the training set.

Our results in Section 8 show that ETS(A,N,A) outperforms ETS(A,A,A) for all cities and all resolutions; that is, including a trend component in ES is detrimental to forecast performance. This can seem surprising, because the time series have a relatively strong trend (Table 3). Trend in a time series can, however, be accounted for, to some extent, by updates to the level component. A disadvantage of including a trend component is that negative forecasts are more likely. To illustrate this, we note that of the forecasts that ETS(A,A,A) produced for Calgary and Edmonton, 0.10%

Fig. 9 Estimated initial seasonal indices of ETS(A,N,A) (left) and TBATS (right)



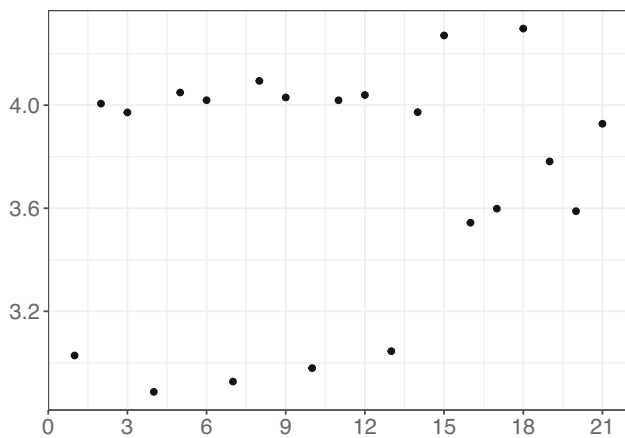


Fig. 10 Seasonal indices of MV

and 0.03% were negative, respectively. Negative forecasts occurred only at the highest spatial resolution (that is, station catchment areas). ETS(A,N,A), TBATS, and MV in contrast, produced no negative forecasts.

An important distinction between TBATS and ETS(A,N,A) is the way in which seasonality is modeled. ETS(A,N,A) estimates 21 initial seasonal indices (one for each 8-hr period of the week), whereas TBATS uses harmonics to estimate seasonal indices. Unlike ETS(A,N,A), the number of parameters TBATS uses to model seasonality is flexible. For long seasonal patterns, TBATS might use fewer parameters than ETS(A,N,A), which could decrease the likelihood of overfitting. Our trained TBATS model uses $k = 9$ harmonics and since two initial parameters are required for each harmonic, TBATS estimated a total of 18 initial parameters. Figure 9 shows the initial seasonal indices for the two methods as points. The aggregate harmonic for TBATS is shown with a dashed curve. (Figure 12 in Appendix C shows the nine individual harmonics.) On the other hand, MV incorporates weekly seasonality through the weekly effect parameters $\hat{\alpha}_1^\omega$ and the $\hat{f}_{i,1}^\omega$ weights used for breaking daily to 8-hr forecasts. Figure 10 shows the “seasonal indices” of MV, calculated as

$\hat{\alpha}_1^\omega \hat{f}_{1,1}^\omega, \hat{\alpha}_1^\omega \hat{f}_{2,1}^\omega, \hat{\alpha}_1^\omega \hat{f}_{3,1}^\omega$, for $\omega = 1, \dots, 7$. We see that they closely resemble the seasonal indices of ETS(A,N,A).

This paper focuses on point forecasts, but distributional forecasts are also important. As we now illustrate, the ETS(A,N,A) model provides a convenient platform for generating distributional forecasts.

Overestimation could have different consequences than underestimation and, therefore, one might wish to use an asymmetric rather than a symmetric loss function, so that an error is penalized not only based on its magnitude but also its sign. If distributional forecasts are available, then one can forecast the quantiles of the call volumes to incorporate an asymmetric loss function [50]. A trained ES method can be used to estimate forecast distributions, by generating sample paths via bootstrapping. We generate sample paths by using the updating equations recursively, using bootstrap samples of the one-step-ahead forecast errors obtained during training to represent the error term, ϵ . Kernel density estimation can then be used to construct bootstrap forecast distributions [40, 6.1]. For ETS(A,N,A), B sample paths of length m (the seasonal cycle length) can be generated as follows:

$$\mathbf{D} = \mathbf{I}_t + \mathbf{E} + \mathbf{s}_t \mathbf{e} + \alpha \mathbf{F},$$

where $\mathbf{D}$ is an $m \times B$ matrix of sample paths that start immediately after Period t , $\mathbf{I}_t$ is the level in Period t , $\mathbf{E}$ is an $m \times B$ matrix of sample errors, $\mathbf{s}_t$ is a column vector containing the most recent seasonal indices for all m periods, as of Period t , $\mathbf{e}$ is a row vector of ones of length B , α is the smoothing parameter for the level, and $\mathbf{F}$ is an $m \times B$ matrix, with $\mathbf{F}_{ij} = \sum_{k=1}^{i-1} \mathbf{E}_{kj}$ for all j and $i \geq 2$, and $\mathbf{F}_{ij} = 0$ for all j and $i = 1$.

Figure 11 shows estimated forecast distributions for 6 horizons obtained using the trained ETS(A,N,A) model, and $B = 5,000$ sample paths. For the MV method, [9] provide a formula to generate forecast distributions that are assumed to be normal distributions. Obtaining corresponding forecast distributions using the TBATS model would require addi-

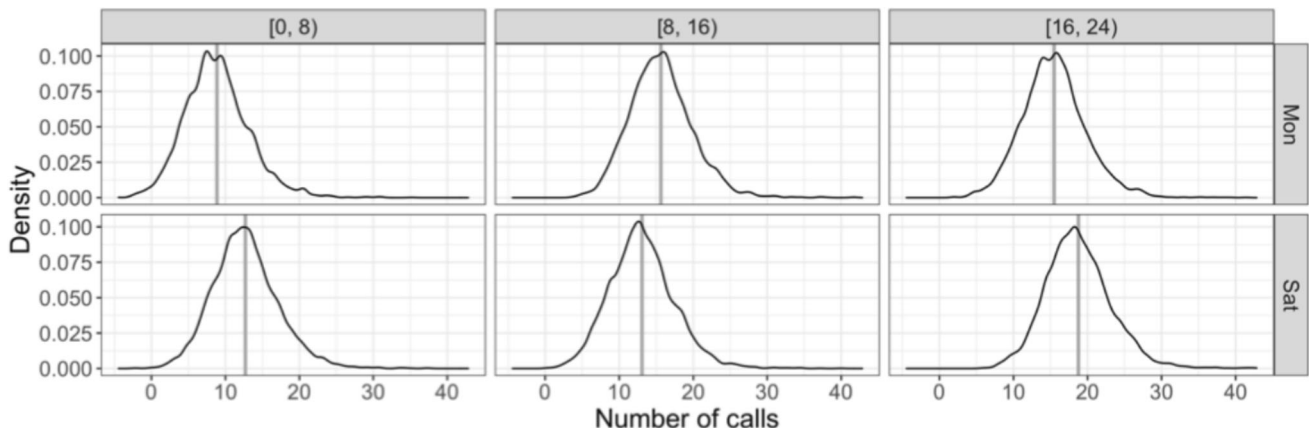


Fig. 11 Forecast distributions of the trained ETS(A,N,A) method for 6 horizons. The vertical line specifies the expected value of the distribution

Table 16 Computation times for Calgary

Method	Runtime (mm:ss)
ETS(A,N,A)	0:41
TBATS	1:24
MV-StoGO	37:10
MV-LBFGS	1:16

tional investigation and the process would likely be more complicated.

Table 16 shows the runtimes of training the ETS(A,N,A), TBATS, and MV methods on the Calgary data at the 8-hr temporal and station catchment area spatial resolution. The results were obtained on an Amazon Web Services EC2 instance featuring a 2nd generation Intel Xeon Platinum 8000 series processor with a clock speed of 3.6GHz, 16 vCPUs, 32GB of RAM, and running Ubuntu 20.04, without parallel computing. Training the MV model involves solving a non-convex optimization problem, specified in Eq. E9, which can be computationally expensive. We provide two results for MV, MV-StoGO and MV-LBFGS, which respectively use the randomized variant of the global optimization algorithm StoGO [51] and the local optimization algorithm Low-storage BFGS [52] to solve Eq. E9. From Table 16 we see that the runtimes of training ETS(A,N,A), TBATS, and MV-LBFGS are less than 1.5 minutes, while the training of MV-StoGO exceeds 37 minutes. Note that the MV-LBFGS might not find a globally optimal solution.

Parallel computation could be used to reduce training time for ETS(A,N,A) and TBATS, because these methods compute forecasts for each region independently. In contrast, it is not clear whether parallel computing could reduce training time for the MV method, because the forecast for each region depends on data from all regions.

10 Discussion

We compared the performance of a variety of ES forecasting methods using 5 years of historical data for three cities in Alberta, Canada. We used three levels of temporal resolution—4-hour, 8-hour, and 24-hour intervals, and three levels of spatial resolution—station catchment areas, intermediate regions, and the entire city.

We found that it is crucial to account for weekly seasonality. Allowing seasonality to be multiplicative rather than additive provided no performance benefit and causes estimation difficulties if the data contains zeros. Other factors, namely: the level of spatio-temporal resolution used for estimation, whether to include trend, whether to use a Box-Cox transformation, and whether to include ARMA error terms have only a minor impact on performance. We found that a simple ES method, having only a level component, estimated

using 24-hour periods and city-level call volumes and disaggregated to 8-hour periods and station catchment areas using a top-down approach performed almost as well as the more complicated TBATS method and outperformed the multivariate MV method and the industry-standard MEDIC method. Our findings were consistent for the three cities.

In some cities, ambulances are deployed dynamically from locations other than stations. To support such deployment, a higher spatial resolution than station catchment areas could be useful. The methods we have discussed could be used in this setting, if one defines catchment areas for every possible ambulance deployment location. We leave an examination of this approach for future studies. Furthermore, in this paper, we focused on point forecasts. We acknowledge the importance of distributional forecasts for EMS planning, as it can assist managers in, for example, quantifying the likelihood of under- or over-estimation of the demand, and we illustrated how the ES methods we used could be used to generate distributional forecasts.

All of the ES methods out-performed a naive method that forecasts the call volume for a future period as the call volume observed one week earlier, as well as the MEDIC moving-average method. Using a simulation study, we found that the performance advantage of a hierarchically reconciled bottom-up approach over a top-down approach increases as the weekly pattern for the central business district diverges from the pattern for the whole city, in terms of amplitude or phase. The advantage for the bottom-up approach is seen only for extreme divergence, such as a 6-hour phase shift for the weekly pattern in the central business district.

The ETS(A,N,A) and the TBATS methods consistently outperformed other methods, for all cities and all spatio-temporal resolutions. These two methods, although similar in terms of performance, differ greatly in terms of their level of complexity. *Simple forecasting*, according to [53], consists of “processes that are understandable to forecast users” and that involve “simple relationships among the model elements”. By [53]’s definition, ETS(A,N,A) is considerably simpler than TBATS. Our findings confirm [53]’s that added complexity does not improve forecasting performance. In addition, simple methods, like ETS(A,N,A), are easier to comprehend and explain and, therefore, are more likely to be used in practice. They are more reliable, because detecting errors is easier if relationships between components of a method are clear and simple. Finally, it is easier to build upon simpler methods, as we demonstrated by constructing distributional forecasts using ETS(A,N,A).

Appendix A: Perturbing the call locations

The geographical location of each call is anonymized by randomly perturbing the latitude and longitude (ϕ , λ) to

$(\phi + \epsilon_1, \lambda + \epsilon_2)$. The perturbation (ϵ_1, ϵ_2) follows a bivariate normal distribution, where the two components have mean zero and zero correlation. The variances are chosen so that the average perturbation area is roughly 400 square metres. The square perturbation distance is approximately:

$$\left(\frac{\pi}{180}R\epsilon_1\right)^2 + \left(\frac{\pi}{180}R\cos(\phi)\epsilon_2\right)^2$$

where R is the Earth's radius and ϕ is the latitude of the city. The approximation is because we are assuming the traversed distance is a straight line. In order for the average area of the perturbed location from the original location to equal 400 square metres, we need that:

$$E\left[\left(\frac{\pi}{180}R\epsilon_1\right)^2 + \left(\frac{\pi}{180}R\cos(\phi)\epsilon_2\right)^2\right]\pi = 0.4$$

Substituting the latitude of Edmonton, $\phi = 53.54$ degrees and $R = 6,371$ km we obtain:

$$\sigma_{\epsilon_1}^2 + 0.3531\sigma_{\epsilon_2}^2 = 1.030 \times 10^{-5}$$

A one degree increase in latitude translates to a larger displacement than a one degree increase in longitude by a factor of $1/\cos(\phi)$. We, therefore, rescale the standard deviations of the perturbation as:

$$\sigma_{\epsilon_1} = \cos(\phi)\sigma_{\epsilon_2}$$

Solving for σ_{ϵ_1} and σ_{ϵ_2} we get $\sigma_{\epsilon_1} = 2.269 \times 10^{-3}$ and $\sigma_{\epsilon_2} = 3.819 \times 10^{-3}$.

Appendix B: Results for methods with multiplicative and damped components

Table 17 shows $\overline{\text{MAE}}$ and $\overline{\text{MASE}}$ values for ES methods with additive and multiplicative trend and seasonality components and damped trend. We cannot train models with multiplicative trend or seasonality if the data includes zero values and therefore we use city-level spatial resolution for training. We use a top-down approach to convert to the catchment area spatial resolution. The temporal resolution is 8 hours. We do not include multiplicative components for Grande Prairie because the data contains zeros even at the city level.

The important pattern that we see in Table 17 is that using multiplicative trend or seasonality, or damped trend, does not provide an advantage over methods that use only additive components.

The results in Table 17 are similar to those in Tables 9–11, for methods with additive components. The slight differences between Table 17 and Tables 9–11 are because of differences in the spatial resolution used for estimation.

Appendix C: TBATS harmonics

Table 17 Performance of methods with different components

Method	Trend	Seasonality	Calgary		Edmonton		Grande Prairie	
			$\overline{\text{MAE}}$	$\overline{\text{MASE}}$	$\overline{\text{MAE}}$	$\overline{\text{MASE}}$	$\overline{\text{MAE}}$	$\overline{\text{MASE}}$
ES	N	N	1.62	88.7%	2.10	89.2%	1.88	82.8%
ES	A	N	1.62	89.1%	2.11	89.6%	1.89	83.0%
ES	Ad	N	1.62	88.8%	2.10	89.3%	1.88	82.8%
ES	M	N	1.62	89.1%	2.11	89.6%		
ES	Md	N	1.62	88.9%	2.10	89.4%		
ES	N	A	1.44	78.4%	1.83	76.7%	1.72	74.3%
ES	N	M	1.44	78.3%	1.83	76.6%		
ES	A	A	1.44	78.4%	1.83	76.7%	1.75	75.5%
ES	Ad	A	1.44	78.4%	1.83	76.7%	1.72	74.5%
ES	A	M	1.44	78.4%	1.83	76.6%		
ES	Ad	M	1.44	78.3%	1.83	76.6%		
ES	M	M	1.44	78.4%	1.83	76.6%		
ES	M	A	1.44	78.4%	1.83	76.7%		
ES	Md	M	1.44	78.3%	1.83	76.6%		
ES	Md	A	1.44	78.4%	1.83	76.7%		
TBATS	Ad	A	1.44	78.1%	1.82	76.2%	1.72	74.3%

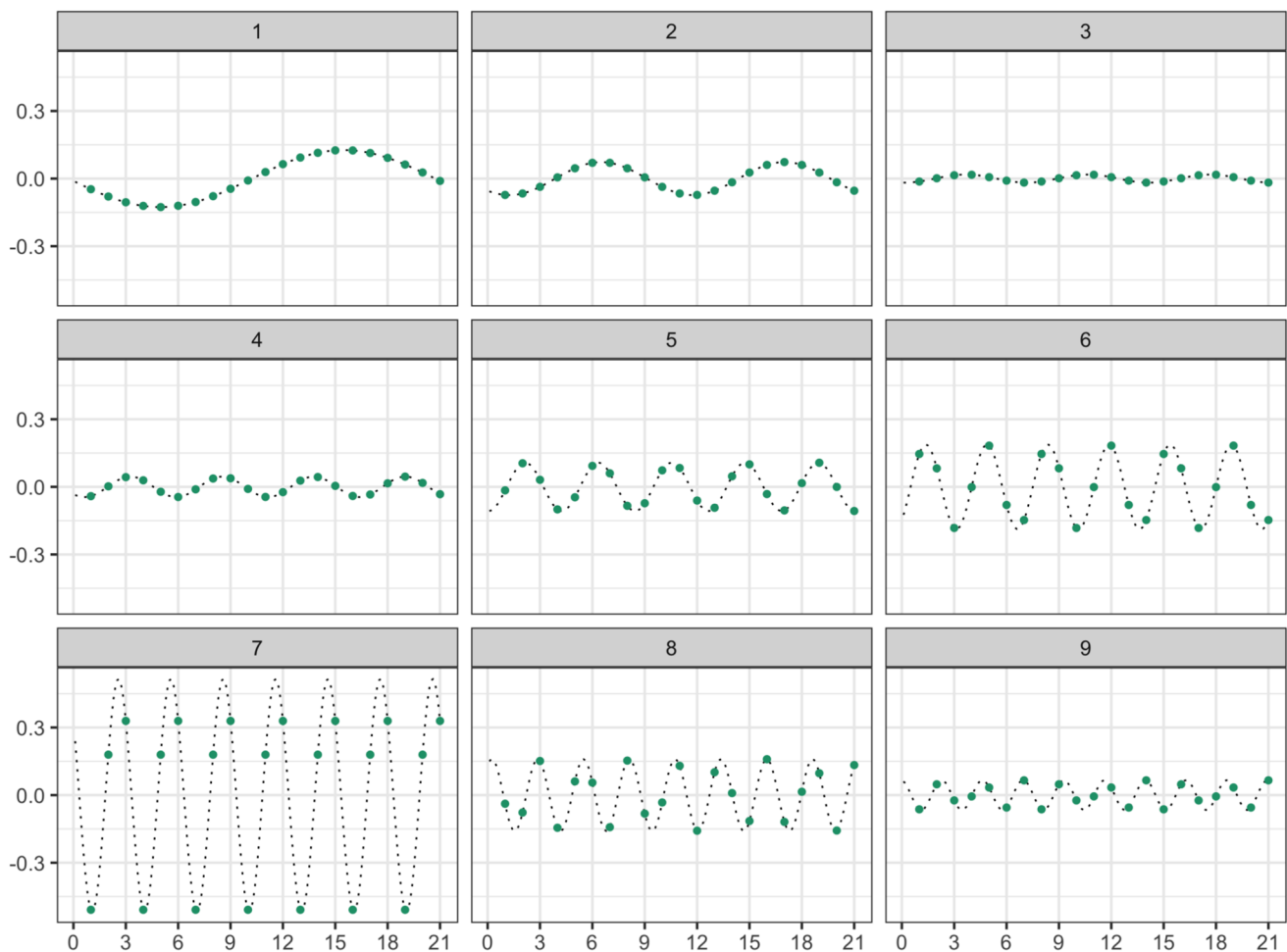


Fig. 12 The $k = 9$ harmonics for the trained TBATS model discussed in Section 9. Points show the seasonal components for each period and dashed curves show the underlying trigonometric functions

Appendix D: Test for spatial correlation

Multiple time series can be forecast either separately or jointly (that is, using either univariate methods or a multivariate method). A call-center forecasting study [9] showed joint forecasting to be beneficial for time series that exhibited inter-stream lag dependence. We test for such dependence, both with and without a lag, among call volumes in intermediate regions ($\alpha = M$). The test procedure proposed by [9], using our notation, is as follows:

1. Calculate the daily square-root transformed call volume of Region r : $q_{t,r} = \sqrt{y_{t,r}^{24,M}}$
2. Define the set $p_d = \{t : \omega(t) = d\}$ of periods corresponding to Day d of the week and calculate its average transformed call volume: $\hat{\beta}_{d,r} = \frac{1}{|p_d|} \sum_{t \in p_d} q_{t,r}$, $d = 1, \dots, 7$, where $|p_d|$ is the cardinality of p_d .
3. Define the vectors $\mathbf{q}_t = (q_{t,r}, r \in R(M))$ and $\hat{\boldsymbol{\beta}}_t = (\hat{\beta}_{\omega(t),r}, r \in R(M))$.

4. Fit the following multivariate time series model using the vars R package [54] and estimate the coefficient matrix $\mathbf{D}$, and the variance-covariance matrix $\tilde{\boldsymbol{\Omega}}$:

$$\mathbf{q}_t - \hat{\boldsymbol{\beta}}_t = \mathbf{D}(\mathbf{q}_{t-1} - \hat{\boldsymbol{\beta}}_{t-1}) + \mathbf{v}_t, \quad \mathbf{v}_t \sim N(0, \tilde{\boldsymbol{\Omega}}) \quad (\text{D1})$$

5. Perform a hypothesis test for the parameters in the coefficient matrix $\mathbf{D} = (d_{ij})$ [55].

A statistically significant coefficient d_{ij} indicates the existence of inter-stream one-day lag dependence of Region i on Region j . Tables 18–19 show the estimated $\mathbf{D}$ matrices for Calgary and Edmonton. Most coefficients are statistically significant (19 out of 25 for Calgary, 15 out of 16 for Edmonton) at the 0.05 level, Bonferroni corrected. All estimated coefficients are positive, indicating that the call volume on Day t in Region r has a positive association with the call volumes on Day $t - 1$ in most of the regions in the city. We performed a multivariate Jarque-Bera test and multivariate

Table 18 Estimated d_{ij} coefficients for Calgary— $\alpha = M$

Region 1	Region 2	Region 3	Region 4	Region 5	
Region 1	0.267*	0.094*	0.018	0.082*	0.079
Region 2	0.075*	0.076*	0.076*	0.109*	0.095*
Region 3	0.042	0.046	0.088*	0.104*	0.114*
Region 4	0.048	0.088*	0.119*	0.153*	0.121*
Region 5	0.087*	0.082*	0.059	0.138*	0.170*

* $p < 0.002$ (Bonferroni adjusted significance level)

skewness and kurtosis tests [56] for normality of the residuals in Model D1. The normality assumption was not rejected.

We performed the same test for the high spatial resolution level (station catchment areas; detailed results omitted). The number of statistically significant coefficients at this spatial resolution was much lower, with only 17 out of 729 for Calgary and 8 out of 256 for Edmonton coefficients being significant at the 0.05 Bonferroni-adjusted level. All significant coefficients were positive, and about half (8 out of 17 for Calgary, 4 out of 8 for Edmonton) corresponded to positive association between call volumes of consecutive days for a specific station catchment area (see Fig. 13).

We further test for spatial correlation among call volumes, that is, correlations among call volumes in different regions, with no lag, and whether such correlations are stronger for regions that are closer to each other. We use the estimated covariance matrix $\hat{\Sigma}$ of the residuals in Eq. D1 to obtain a matrix of estimated correlations, (ρ_{ij}) . Figure 14 shows how the ρ_{ij} depend on the distances between station catchment areas i and j . We fit piece-wise linear models to the correlations, using 5 km intervals. For Calgary, we only obtain a significant estimate for the distance coefficient in the (0, 5km] interval, with an estimated coefficient of -0.02 (p -value < 0.001), whereas for Edmonton we do not obtain any significant coefficients. Thus, we do not find strong evidence for a spatial correlation effect.

Appendix E: MV model implementation

In this section, we provide the details of implementing the multivariate forecasting model of [9], MV. The code for train-

Table 19 Estimated d_{ij} coefficients for Edmonton— $\alpha = M$

Region 1	Region 2	Region 3	Region 4	
Region 1	0.230*	0.078*	0.080*	0.091*
Region 2	0.074*	0.195*	0.105*	0.157*
Region 3	0.069*	0.068*	0.075*	0.105*
Region 4	0.106*	0.177*	0.049	0.200*

* $p < 0.002$ (Bonferroni adjusted significance level)

ing the model and obtaining forecasts can be accessed from <https://github.com/mrez746/Multivariate-forecasting>.

We follow the procedure in Section 3.2 of [9] for training Model (8) but we elaborate on how we implemented certain steps.

The authors of [9] provide an iterative procedure for estimating the parameters of their proposed method. Step 2(a) requires fitting the following GLS model for obtaining estimates of the $f_{r,i(t)}^{\omega(t)}$ parameters, given estimates of $u_{r,d(t)}$ and Σ :

$$x_{t,r} = \hat{u}_{d(t),r} f_{i(t),r}^{\omega(t)} + \epsilon_{t,r}, \quad (\text{E2})$$

$$\epsilon_t = (\epsilon_{t,1}, \epsilon_{t,2}, \dots, \epsilon_{t,m}) \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \hat{\Sigma}),$$

$$\sum_i f_{i,r}^{\omega} = 1, \text{ for } r = 1, \dots, m, \omega = 1, \dots, 7.$$

We write Model (E2) in matrix format to facilitate solving the constrained GLS using CVXR, a package for specifying and solving convex programs [57]. Let $\hat{\mathbf{U}}$ denote a matrix holding estimates of $u_{d,r}$, for $r = 1, \dots, m$ and $d = 1, \dots, D$, and $\mathbf{f}$ a vector holding estimates of $f_{i,r}^{\omega}$ for $r = 1, \dots, m, i = 1, \dots, I$, and $\omega = 1, \dots, 7$. In this step, we keep $\hat{\mathbf{U}}$ fixed but estimate the values in $\mathbf{f}$. The first two lines of Model (E2) can be written in matrix notation as

$$\mathbf{x} = \hat{\mathbf{U}}\mathbf{f} + \boldsymbol{\epsilon}, \quad (\text{E3})$$

with

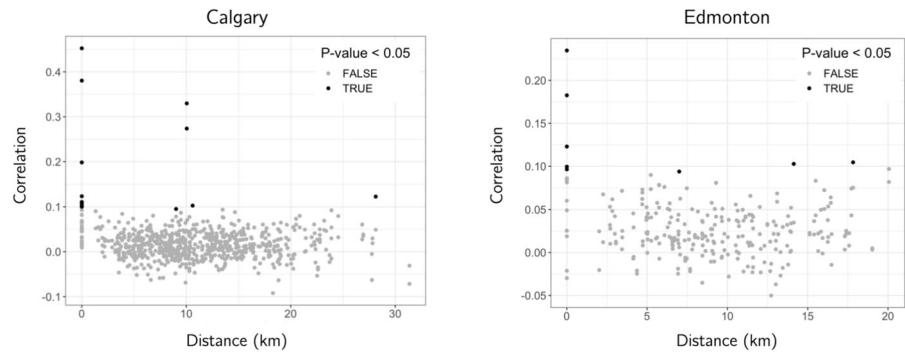
- $\mathbf{x}$ a length mDI column vector holding $x_{t,r}$ in its $(r-1)DI + (d(t)-1)I + i(t)$ position. for $r = 1, \dots, m, t = 1, \dots, DI$;
- $\hat{\mathbf{U}}$ a mDI by $7mI$ matrix holding $\hat{u}_{d,r}$ in its Row $(r-1)DI + (d-1)I + i$, Column $7(r-1)I + (\omega(d)-1)I + i$ position, for $r = 1, \dots, m, d = 1, \dots, D, i = 1, \dots, I$, and 0 elsewhere. ($\hat{\mathbf{U}}$, therefore, contains i copies of $\hat{u}_{d,r}$.)
- $\mathbf{f}$ a length $7mI$ column vector holding $f_{i,r}^{\omega}$ in its $7(r-1)I + (\omega-1)I + i$ position. for $\omega = 1, \dots, 7, r = 1, \dots, m, i = 1, \dots, I$; and
- $\boldsymbol{\epsilon}$ a length mDI column vector holding $\epsilon_{t,r}$ in its $(r-1)DI + (d(t)-1)I + i(t)$ position. for $r = 1, \dots, m, t = 1, \dots, DI$.

Step 2(c) requires fitting the following GLS model for obtaining estimates of the $u_{d(t),r}$ parameters, given estimates of $f_{i(t),r}^{\omega(t)}$ and Σ :

$$x_{t,r} = \hat{f}_{i(t),r}^{\omega(t)} u_{d(t),r} + \epsilon_{t,r},$$

$$\epsilon_t = (\epsilon_{t,1}, \epsilon_{t,2}, \dots, \epsilon_{t,m}) \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \hat{\Sigma}). \quad (\text{E4})$$

Fig. 13 Estimated d_{ij} coefficients as a function of distance between stations, for $\alpha = H$



Let $\hat{\mathbf{F}}$ denote a matrix holding estimates of $f_{i,r}^\omega$ for $r = 1, \dots, m$, $i = 1, \dots, I$, and $\omega = 1, \dots, 7$, and $\mathbf{u}$ a vector holding estimates of $u_{d,r}$, for $r = 1, \dots, m$ and $d = 1, \dots, D$. In this step, we keep $\hat{\mathbf{F}}$ fixed but estimate the values in $\mathbf{u}$. Model (E4) can be written in matrix notation as

$$\mathbf{x} = \hat{\mathbf{F}}\mathbf{u} + \boldsymbol{\epsilon}, \quad (\text{E5})$$

with

- $\hat{\mathbf{F}}$ a mDI by mD matrix, holding $f_{i,r}^\omega$ in its Row $(r-1)DI + (d-1)I + i$, Column $(r-1)D + d$ position, for $\omega = 1, \dots, 7$, $r = 1, \dots, m$, $i = 1, \dots, I$, $d : \omega(d) = \omega$, and 0 elsewhere. ($\hat{\mathbf{F}}$, therefore, may contain multiple copies of $f_{i,r}^\omega$.)
- $\mathbf{u}$ a column vector of length mD , holding $u_{d,r}$ in its $(r-1)D + d$ position. for $r = 1, \dots, m$, $d = 1, \dots, D$;

and $\mathbf{x}$ and $\boldsymbol{\epsilon}$ as before.

For Calgary at the station catchment area and 8-hr resolution, the length of the vector $\mathbf{f}$ in Eq. E3 (that is, the number of values to be estimated) is $7 \times 27 \times 3 = 567$ and the length of the vector $\mathbf{u}$ in Eq. E5 (that is, the number of values to be estimated) is $27 \times 1,064 = 28,728$.

Once estimates of $u_{d,r}$, $f_{i,r}^\omega$, and $\boldsymbol{\Sigma}$ are obtained, we estimate the remaining parameters following the procedure in Section 3.2.2 of [9], which requires training the following

model

$$\hat{\mathbf{u}}_d - \boldsymbol{\alpha}_{\omega(d)} = \mathbf{A}(\hat{\mathbf{u}}_{d-1} - \boldsymbol{\alpha}_{\omega(d-1)}) + \mathbf{z}_d, \quad \mathbf{z}_d \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \boldsymbol{\Omega}), \quad d = 2, \dots, D. \quad (\text{E6})$$

This is an iterative procedure: we start with initial estimates of $\boldsymbol{\alpha}_{\omega(d)}$; we plug them into Eq. E6 and solve the model to obtain estimates of $\mathbf{A}$ and $\boldsymbol{\Omega}$; we then plug the two new estimates in Eq. E6 to obtain updated estimates of $\boldsymbol{\alpha}_{\omega(d)}$, using MLE. We repeat these steps until all estimates converge. We provide details for the second step (updating the estimates for $\boldsymbol{\alpha}_{\omega(d)}$) below.

Given estimates of $\mathbf{A}$ and $\boldsymbol{\Omega}$, we have

$$\hat{\mathbf{A}}\boldsymbol{\alpha}_{\omega(d-1)} - \boldsymbol{\alpha}_{\omega(d)} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\hat{\mathbf{A}}\hat{\mathbf{u}}_{d-1} - \hat{\mathbf{u}}_d, \hat{\boldsymbol{\Omega}}), \quad d = 2, \dots, D.$$

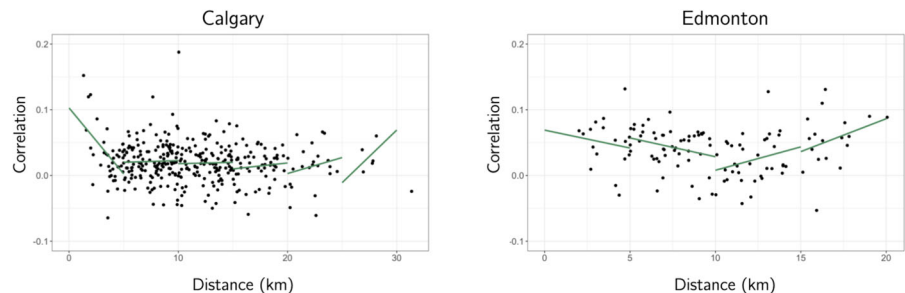
To obtain the maximum-likelihood estimates of $\boldsymbol{\alpha}_{\omega(d)}$, we solve the following optimization problem

$$\max_{\boldsymbol{\alpha}_1, \dots, \boldsymbol{\alpha}_7} \prod_{d=2}^D f_d(\hat{\mathbf{A}}\boldsymbol{\alpha}_{\omega(d-1)} - \boldsymbol{\alpha}_{\omega(d)}), \quad (\text{E7})$$

where f_d is the PDF of the multivariate normal distribution with mean $\boldsymbol{\mu}_d = \hat{\mathbf{A}}\hat{\mathbf{u}}_{d-1} - \hat{\mathbf{u}}_d$ and covariance matrix $\hat{\boldsymbol{\Omega}}$:

$$f_d(\hat{\mathbf{A}}\boldsymbol{\alpha}_{\omega(d-1)} - \boldsymbol{\alpha}_{\omega(d)}) = (2\pi)^{-\frac{I}{2}} |\det(\hat{\boldsymbol{\Omega}})|^{-\frac{1}{2}} \exp \left[-\frac{1}{2} (\hat{\mathbf{A}}\boldsymbol{\alpha}_{\omega(d-1)} - \boldsymbol{\alpha}_{\omega(d)} - \boldsymbol{\mu}_d)^\top \hat{\boldsymbol{\Omega}}^{-1} (\hat{\mathbf{A}}\boldsymbol{\alpha}_{\omega(d-1)} - \boldsymbol{\alpha}_{\omega(d)} - \boldsymbol{\mu}_d) \right].$$

Fig. 14 Estimated ρ_{ij} coefficients as a function of distance between stations, for $\alpha = H$, with piece-wise linear fits



The product in Problem (E7) starts at $d = 2$ because the vector autoregressive model in Eq. E6 starts at $d = 2$. The log-likelihood is thus:

$$\begin{aligned} ll &= \log \prod_{d=2}^D f_d(\hat{\mathbf{A}}\boldsymbol{\alpha}_{\omega(d)-1} - \boldsymbol{\alpha}_{\omega(d)}) \\ &= \sum_{d=2}^D \log \left((2\pi)^{-\frac{1}{2}} |\det(\hat{\boldsymbol{\Omega}})|^{-\frac{1}{2}} \right. \\ &\quad \left. \exp \left[-\frac{1}{2} (\hat{\mathbf{A}}\boldsymbol{\alpha}_{\omega(d)-1} - \boldsymbol{\alpha}_{\omega(d)} - \boldsymbol{\mu}_d)^\top \hat{\boldsymbol{\Omega}}^{-1} (\hat{\mathbf{A}}\boldsymbol{\alpha}_{\omega(d)-1} - \boldsymbol{\alpha}_{\omega(d)} - \boldsymbol{\mu}_d) \right] \right) \\ &= (D-1) \log \left((2\pi)^{-\frac{1}{2}} |\det(\hat{\boldsymbol{\Omega}})|^{-\frac{1}{2}} \right) + \\ &\quad \sum_{d=2}^D \left(-\frac{1}{2} (\hat{\mathbf{A}}\boldsymbol{\alpha}_{\omega(d)-1} - \boldsymbol{\alpha}_{\omega(d)} - \boldsymbol{\mu}_d)^\top \hat{\boldsymbol{\Omega}}^{-1} (\hat{\mathbf{A}}\boldsymbol{\alpha}_{\omega(d)-1} - \boldsymbol{\alpha}_{\omega(d)} - \boldsymbol{\mu}_d) \right) \\ &= c_1 - \frac{1}{2} \sum_{d=2}^D \left((\boldsymbol{\alpha}_{\omega(d)-1}^\top \hat{\mathbf{A}}^\top - \boldsymbol{\alpha}_{\omega(d)}^\top - \boldsymbol{\mu}_d^\top) \hat{\boldsymbol{\Omega}}^{-1} (\hat{\mathbf{A}}\boldsymbol{\alpha}_{\omega(d)-1} - \boldsymbol{\alpha}_{\omega(d)} - \boldsymbol{\mu}_d) \right) \\ &= c_1 - \frac{1}{2} \sum_{d=2}^D \left(\boldsymbol{\alpha}_{\omega(d)-1}^\top \hat{\mathbf{A}}^\top \hat{\boldsymbol{\Omega}}^{-1} \hat{\mathbf{A}}\boldsymbol{\alpha}_{\omega(d)-1} + \boldsymbol{\alpha}_{\omega(d)}^\top \hat{\boldsymbol{\Omega}}^{-1} \boldsymbol{\alpha}_{\omega(d)} + \boldsymbol{\mu}_d^\top \hat{\boldsymbol{\Omega}}^{-1} \boldsymbol{\mu}_d \right. \\ &\quad \left. - 2\boldsymbol{\alpha}_{\omega(d)-1}^\top \hat{\mathbf{A}}^\top \hat{\boldsymbol{\Omega}}^{-1} \boldsymbol{\alpha}_{\omega(d)} - 2\boldsymbol{\alpha}_{\omega(d)-1}^\top \hat{\mathbf{A}}^\top \hat{\boldsymbol{\Omega}}^{-1} \boldsymbol{\mu}_d + 2\boldsymbol{\alpha}_{\omega(d)}^\top \hat{\boldsymbol{\Omega}}^{-1} \boldsymbol{\mu}_d \right), \end{aligned}$$

where $c_1 = (D-1) \log \left((2\pi)^{-\frac{1}{2}} |\det(\hat{\boldsymbol{\Omega}})|^{-\frac{1}{2}} \right)$.

Removing terms that do not depend on $\boldsymbol{\alpha}_\omega$, for $\omega = 1, \dots, 7$ we obtain the following problem, which has the same solution as Problem (E7)

$$\min_{\boldsymbol{\alpha}_1, \dots, \boldsymbol{\alpha}_7} \sum_{d=2}^D \left(\boldsymbol{\alpha}_{\omega(d)-1}^\top \hat{\mathbf{A}}^\top \hat{\boldsymbol{\Omega}}^{-1} \hat{\mathbf{A}}\boldsymbol{\alpha}_{\omega(d)-1} + \boldsymbol{\alpha}_{\omega(d)}^\top \hat{\boldsymbol{\Omega}}^{-1} \boldsymbol{\alpha}_{\omega(d)} \right. \quad (\text{E8}) \\ \left. - 2\boldsymbol{\alpha}_{\omega(d)-1}^\top \hat{\mathbf{A}}^\top \hat{\boldsymbol{\Omega}}^{-1} \boldsymbol{\alpha}_{\omega(d)} - 2\boldsymbol{\alpha}_{\omega(d)-1}^\top \hat{\mathbf{A}}^\top \hat{\boldsymbol{\Omega}}^{-1} \boldsymbol{\mu}_d + 2\boldsymbol{\alpha}_{\omega(d)}^\top \hat{\boldsymbol{\Omega}}^{-1} \boldsymbol{\mu}_d \right).$$

Of the five terms inside the summation, the second term is a convex (because $\hat{\boldsymbol{\Omega}}^{-1}$ is positive definite) and the last two terms are convex (because they are linear). The first and the third term, however, are not guaranteed to be convex, and therefore this minimization problem is not guaranteed to be convex.

Problem (E8) can be written as

$$\min_{\boldsymbol{\alpha}_1, \dots, \boldsymbol{\alpha}_7} \sum_{j=1}^7 \left(n_j^{(1)} \boldsymbol{\alpha}_j^\top \hat{\mathbf{A}}^\top \hat{\boldsymbol{\Omega}}^{-1} \hat{\mathbf{A}}\boldsymbol{\alpha}_j + n_j^{(2)} \boldsymbol{\alpha}_j^\top \hat{\boldsymbol{\Omega}}^{-1} \boldsymbol{\alpha}_j \right. \quad (\text{E9}) \\ \left. - 2n_j^{(2)} \boldsymbol{\alpha}_{j-1}^\top \hat{\mathbf{A}}^\top \hat{\boldsymbol{\Omega}}^{-1} \boldsymbol{\alpha}_j - 2\boldsymbol{\alpha}_{j-1}^\top \hat{\mathbf{A}}^\top \hat{\boldsymbol{\Omega}}^{-1} \mathbf{v}_j \mathbf{e}_j + 2\boldsymbol{\alpha}_j^\top \hat{\boldsymbol{\Omega}}^{-1} \mathbf{v}_j \mathbf{e}_j \right),$$

where $n_j^{(1)} = \sum_{d=1}^{D-1} \mathbb{1}(\omega(d) = j)$, $n_j^{(2)} = \sum_{d=2}^D \mathbb{1}(\omega(d) = j)$, $\mathbf{v}_j$ is a i by $n_j^{(2)}$ matrix with $\boldsymbol{\mu}_d : \omega(d) = j$, for $d = 2, \dots, D$ as its columns, $\mathbf{e}_j$ is a length $n_j^{(2)}$ column vector of ones, and $\boldsymbol{\alpha}_0 = \boldsymbol{\alpha}_7$.

The partial derivative of the objective function of Problem (E9) with respect to $\boldsymbol{\alpha}_j$ is

$$\begin{aligned} &\left(2n_j^{(1)} \hat{\mathbf{A}}^\top \hat{\boldsymbol{\Omega}}^{-1} \hat{\mathbf{A}} + n_j^{(2)} \hat{\boldsymbol{\Omega}}^{-1} \right) \boldsymbol{\alpha}_j - \left(2n_j^{(2)} (\hat{\boldsymbol{\Omega}}^{-1})^\top \hat{\mathbf{A}} \right) \boldsymbol{\alpha}_{j-1} - \\ &\left(2n_{j+1}^{(2)} \hat{\mathbf{A}}^\top \hat{\boldsymbol{\Omega}}^{-1} \right) \boldsymbol{\alpha}_{j+1} + 2\hat{\boldsymbol{\Omega}}^{-1} \mathbf{v}_j \mathbf{e}_j - 2\hat{\mathbf{A}}^\top \hat{\boldsymbol{\Omega}}^{-1} \mathbf{v}_{j+1} \mathbf{e}_{j+1}, \end{aligned}$$

with $\boldsymbol{\alpha}_8 = \boldsymbol{\alpha}_1$, $\mathbf{v}_8 = \mathbf{v}_1$, and $\mathbf{e}_8 = \mathbf{e}_1$.

We solve Problem (E9) by calculating the gradient of the objective function and using the global optimization algorithm StoGO [51], using the R package nloptr [58]. For the optimization routine's starting point and bound constraints, we first define

$$\begin{aligned} \hat{\boldsymbol{\alpha}}_\omega^{(0)} &= \frac{1}{n_\omega} \sum_{d:\omega(d)=\omega} \hat{\mathbf{u}}_d, \text{ for } \omega = 1, \dots, 7, \\ s_\omega^2 &= \frac{1}{n_\omega - 1} \sum_{d:\omega(d)=\omega} (\hat{\mathbf{u}}_d - \hat{\boldsymbol{\alpha}}_\omega^{(0)})^2, \end{aligned}$$

with $n_\omega = \sum_{d=1}^D \mathbb{1}\{\omega(d) = \omega\}$, for $\omega = 1, \dots, 7$. For Iteration i of the procedure in Section 3.2.2 of [9], we set the starting point to $\hat{\boldsymbol{\alpha}}_\omega^{(i-1)}$ and the boundary constraint to $[\hat{\boldsymbol{\alpha}}_\omega^{(i-1)} - s_\omega, \hat{\boldsymbol{\alpha}}_\omega^{(i-1)} + s_\omega]$, with $\hat{\boldsymbol{\alpha}}_\omega^{(i)}$ the solution obtained by StoGO in Iteration i . We use the randomizing variant of StoGO and set the maximum iteration and tolerance of the algorithm to 10,000 and 10^{-6} , respectively.

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Declarations

This study was funded by the Canadian Natural Science and Engineering Research Council under Grant Agreement No. DG 203534. The authors have no relevant financial or non-financial interests to disclose. Ethics approval was obtained from the University of Alberta Research Ethics Board (identification number Pro00071718). We provide code developed by us for training one of the benchmark methods, at <https://github.com/mrez746/Multivariate-forecasting>. We used publicly available R packages for training all other methods.

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